# TASK-RELEVANT SHIFTS IN CATEGORY REPRESENTATIONS

BY

MATT WETZEL

B*S*, Cleveland State University, 2014

THESIS

Submitted in partial fulfillment of the requirements for
the degree of Master of Science in Psychology
in the Graduate School of
Binghamton University
State University of New York
2020

© Copyright by Matt Wetzel 2020

All Rights Reserved

Accepted in partial fulfillment of the requirements for
the degree of Master of Science in Psychology
in the Graduate School of
Binghamton University
State University of New York
2020

January 24, 2020

Dr. Kenneth J. Kurtz, Chair
Department of Psychology, Binghamton University

Dr. Deanne L. Westerman, Member
Department of Psychology, Binghamton University

Dr. Peter C. Gerhardstein, Member
Department of Psychology, Binghamton University

# Abstract

Categorical perception is a phenomenon where engagement in a category learning task leads to changes in the reported similarity of objects. This often manifests as an increase in within-category similarity ratings and a decrease in between-category similarity ratings relative to a control. In the domain of artificial category learning, categorical perception effects have typically been observed using a classification learning paradigm, where learners' task is to identify the category label of a stimulus given a set of options. The present work explores categorical perception effects in the context of endorsement learning, where learners' task is to determine whether a provided category label matches a stimulus. The results suggest that categorical perception effects, as typically observed in classification learning, also occur during endorsement learning. Leading computational models of category learning were assessed according to their ability to predict the present results.

# Contents

**List of Tables** **vi**

**List of Figures** **vii**

**1 Introduction** **1**
1.1 Foundations of the Traditional Artificial Category Learning Paradigm (TACL) . . . 2
1.2 Theories of Categorization . . . 5
1.3 Categorical Perception . . . 7
1.4 Connectionism and Categorical Perception . . . 11
1.5 Limitations of TACL and Alternatives to Classification Learning . . 16
1.6 The Present Work . . . 19

**2 Behavioral Experiment** **21**
2.1 Methods . . . 21
2.1.1 Participants . . . 21
2.1.2 Design . . . 21
2.1.3 Materials . . . 22
2.1.4 Procedure . . . 23
2.2 Results & Discussion . . . 24
2.2.1 Learning Difficulty . . . 24
2.2.2 Similarity Ratings . . . 27
2.2.3 Post-Hoc Analyses . . . 29

**3 Computational Experiments** **36**
3.1 Candidate Models . . . 37
3.1.1 Attention Learning COVEring map . . . 37
3.1.2 DIVergent Autoencoder . . . 39
3.1.3 Theory Of Task . . . 43
3.2 Simulation Methods . . . 45
3.2.1 Assessing Psychological Plausibility . . . 46
3.2.2 Querying Representational Space . . . 48
3.3 Results & Discussion . . . 50

**4 General Discussion** **55**

**References** **64**

# List of Tables

2.1 Distribution of Subjects . . . . . 21

3.1 Simulation Results . . . . . 51
3.2 Best Fitting Hyperparameters - Classification . . . . . 51
3.3 Best Fitting Hyperparameters - Endorsement . . . . . 52

# List of Figures


1.1 Shepard's Law . . . . . 5
1.2 Representational Similarity Matrix . . . . . 15
1.3 Learning Modes . . . . . 19

2.1 Category Structures . . . . . 23
2.2 First Perfect Block Scores & Learning Curves . . . . . 26
2.3 Endorsement Feedback Scenarios . . . . . 26
2.4 Similarity Ratings for Each Learning Mode . . . . . 29
2.5 Graphical Representation of Similarity Rating Data . . . . . 30
2.6 Aggregated Generalization Profiles . . . . . 31
2.7 Attention Profiles . . . . . 32
2.8 Between-Category Expansion as a Function of Pair Distance . . . . 34

3.1 Impact of ALCOVE's attention on feature space . . . . . 39
3.2 Autoencoder Representational Space Over Time . . . . . 53
3.3 Hidden Representational Space . . . . . 54
3.4 Calculating Fits to Learning Data . . . . . 54

4.1 Category Structures for Exploring Categorical Perception in ALCOVE . . . . . 62
4.2 Category Structures for Exploring Categorical Perception in DIVA . 63

# 1 Introduction

Human beings are relatively adept at recognizing statistical patterns in their environment and generalizing those patterns to novel experiences. For example, it is relatively easy to drive a great variety of cars after only learning to drive one. In some sense, we can think of this process as "categorization", where prior knowledge is selectively generalized to new and similar contexts. Cognitive scientists have had a long standing interest in how categorization impacts the way humans *behave* in their environment. In addition, cognitive scientists have begun to explore the question of how categorization impacts the way humans *experience* their environment as well.

Early work on categorization explored the the philosophical question of defining what categories are. The "classic view" of categorization posits that objects/concepts belong to a category if they have all of the *necessary and sufficient* properties of that category (Murphy, 2004; Smith & Medin, 1981). While the classical view of categorization is logically elegant, it has been heavily criticized for its lack of ecological validity: it is difficult to find real world categories that can be neatly defined by a set of necessary and sufficient properties (Wittgenstein, 1953).

Another key limitation of the classic view of categorization is the assump-

tion that category membership is binary[1]. An alternative is to assume a more "graded" view of category membership. Evidence of this view comes from empirical work showing that human subjects report that some objects/concepts are "better" members of a category than others (Rosch, 1975). Rosch and Mervis (1975) suggest that many natural categories encompass some "family-resemblance" between all the categories' members; under this view, something is a member of a category if it closely resembles the *prototypical* properties of the other members of the category. This theoretical view and the many theories that followed assume that categories emerge from the statistical properties of their members. There have since been a number of different theoretical accounts of how humans might learn and generalize the distributional properties of categories (Nosofsky, 1986; Ashby & Lee, 1991; Kruschke, 1992; Kurtz, 2007; Love, Medin, & Gureckis, 2004; Anderson, 1991)

## 1.1 Foundations of the Traditional Artificial Category Learning Paradigm (TACL)

There are a number of ways in which humans could learn categorical information in a laboratory settings (Love et al., 2004; Markman & Ross, 2003; Levering & Kurtz, 2015). An ideal paradigm would allow researchers to tightly control the learning experience in order to test competing theoretical predictions about how humans form categorical representations. That is, an ideal paradigm for studying category learning would:

- use stimuli whose features are quantitatively describable (so that the statis-

[1] i.e., something is in a category or it isn't

tical properties of their constituent categories are apparent and manipulatable)

- use stimuli whose features are easily identifiable (so that subjects know what to pay attention to)
- involve domains that are ecologically atypical (so that the only knowledge subjects can utilize are the statistical properties of the experimental stimuli)

This type of paradigm was utilized by Shepard, Hovland, and Jenkins (1961), who "trained" human learners to correctly identify arbitrary labels assigned to different groups of abstract stimuli; each stimulus was represented as a set of quantitatively-describable features (e.g., size, color, shape). Subjects learned how to map stimuli to labels through trial and error. On each trial of learning, subjects made a guess about the category membership of a stimulus. Following their guess, they were given corrective feedback from the experimenter. Overtime, their ability to identify the category of stimuli improved. This fundamental paradigm has been extensively utilized by category theorists for decades (Nosofsky, 1986; Kurtz, 2007; Anderson, 1991; Shepard et al., 1961; Bruner, Goodnow, & Austin, 1956). Importantly, it has also served as an empirical benchmark for developing computational theories of human categorization as well (Wills & Pothos, 2012; Kurtz, 2015).

The TACL paradigm has shed light on a number of important properties about human category learning of artificial stimuli. One key finding from Shepard et al.'s (1961) seminal work was that simple, abstract categories are easier to discriminate when the number of relevant dimensions is small. In their original work, Shepard et al. (1961) trained subjects on stimuli with 3 binary features; each stimulus

could belong to two possible categories. A well-replicated finding from this work is that categories of this kind are most quickly learned when they are defined by a perfectly predictive, unidimensional rule (or, a decisional "boundary" line in feature space Nosofsky, Gluck, Palmeri, McKinley, & Glauthier, 1994). Follow-up work has found that human learners have difficulty utilizing the predictive value of multiple features during classification learning (preferring instead to focus on one feature at a time; Kruschke, 1993). Consequently, many computational accounts of human category learning utilize some form of featural attention to account for this particular phenomenon (Nosofsky, 1986; Kruschke, 1992; Love et al., 2004; Erickson & Kruschke, 1998)[2].

Another key finding comes from Shepard's (1987) work demonstrating that human generalization can often be predicted by an exponentially decaying distance function. That is, the likelihood that a subject will generalize a target stimulus to a category is given by the similarity between the target stimulus and some reference point that represents that category in some geometric space (see figure 1.1). Importantly, Shepard's (1987) "law" of generalization also describes a number of other psychological phenomena, such as neural tuning to sensory information (Kang, Shapley, & Sompolinsky, 2004; Butts & Goldman, 2006) and response generalization to reinforcing stimuli (Guttman & Kalish, 1956). Shepard's (1987) findings on stimulus generalization have been the inspiration behind a number of computational theories of human category learning within the TACL framework (Nosofsky, 1986; Kruschke, 1992; Minda & Smith, 2001).

[2]This phenomenon might reflect a similar finding in the reinforcement literature, where researchers have found that reinforcement models provide better accounts of animal learning when they restrict models to learning one-to-one associations between cues and outcomes, as opposed to many-to-one associations (Witnauer, Urcelay, & Miller, 2014)

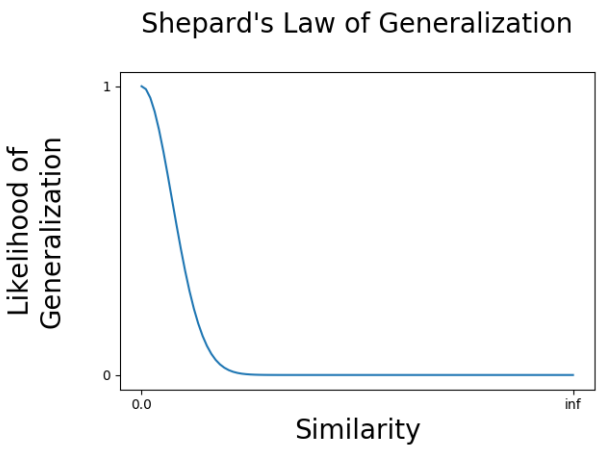


Figure 1.1: Shepard's Law can be represented as a monotonic, inverse exponential distance function (Shepard, 1987) giving the likelihood of generalizing the learned representation of a stimulus to a new, unexperienced stimulus.

## 1.2 Theories of Categorization

Shepard's (1987) law of generalization suggests that new target stimuli are categorized based on their exponentially-decaying similarity to reference points in memory. A great deal of debate has ensued around what those reference points are. Some theories postulate that new stimuli are compared to a description of the average properties of all previously experienced stimuli; this is typically referred to as the "prototype" view (Minda & Smith, 2001). Other theories postulate that new stimuli are categorized according to their individual similarity to each unique stimulus in memory; this is typically referred to as the "exemplar" view (Nosofsky, 1986). These frameworks (and their many variants) have been foundational to cognitive psychology, and the theoretical debates that surround them have lead to a number of important findings about how people behave in TACL experiments. Exemplar models have had notable successes in predicting the learning and generalization behavior of humans (Kruschke, 1992; Nosofsky, Gluck, et al., 1994; Palmeri, 1999; Nosofsky & Johansen, 2000), supporting Shepard's (1987) account

of stimulus generalization as well as the idea that category representations are composed of aggregated reference points in memory.

Another theoretical account assumes that category representations consist of dimensional boundaries that optimally separate categories in featural space. In some accounts, these boundaries are learned by iteratively testing diagnostic "rules" that describe a conditional relationship[3] between category membership and a restricted set of features Nosofsky and Palmeri (1998). In other variants, boundaries are optimized to maximally separate inferred category distributions through which exemplars are sampled Ashby and Gott (1988). Other theoretical approaches incorporate a hybrid of both reference point mechanisms and dimensional boundaries (Erickson & Kruschke, 1998; Ashby, Alfonso-Reese, Waldron, et al., 1998).

Many formal models suggest that classification decisions are made by computing the likelihood of a category label given a provided set of stimulus features. Alternatively, classification decisions could be the result of determining the likelihood of a set of features given a candidate category (Levering & Kurtz, 2015). Under this framework, categorization reflects a process of building a statistical model that best describes the categories the learner experiences. Once a statistical model is learned, classification decisions are made by optimally deciding which category model best describes a presented stimulus (Kurtz, 2007; Anderson, 1991).

The psychological plausibility of any particular theoretical model has traditionally been assessed according to how well it predicts human behavior (Wills & Pothos, 2012). "Behavior" is typically measured by testing (a) the ease in which categories are learned, and (b) the way learners generalize category labels to new

[3]biasing conditional relationships that are perfectly predictive

stimuli (Kurtz, 2015). Many existing models have provided successful accounts of both of these metrics (Wills & Pothos, 2012). However, many researchers have also been interested in how category representations shape our experience of the world around us; that is, how category learning influences our *perception*.

## 1.3 Categorical Perception

The process through which categories shape our experience is commonly referred to as "categorical perception" (Livingston, Andrews, & Harnad, 1998; Goldstone, 1994a). Categorical perception has been studied in a variety of contexts. Speech perception has been a particularly fruitful research domain through which categorical perception has been explored. Speech perception is an ideal domain for studying categorical perception since spoken languages have an inherently categorical structure. For example, researchers have found that it is much easier for humans to discriminate speech sounds when they vary along phonemic categories. In one groundbreaking study, Liberman, Harris, Hoffman, and Griffith (1957) created a set of synthetic sounds that varied along continuous acoustic dimensions. Human learners were trained to classify these stimuli into labeled categories provided by the experimenters. Importantly, Liberman et al. (1957) found that human learners could more easily discriminate 2 sounds when they belonged to different phonemic categories (relative to when they belonged to the same category). Interestingly, evidence of category perception during the processing of phonemes has been found in infants, and may reflect an innate computational efficiency of forming discrete descriptions of continuous signals (Harnad, 1987).

Importantly, categorical perception has also been studied under the TACL

framework. In one foundational study, Goldstone (1994b) found strong evidence for categorical perception in a TACL experiment using 2 dimensional stimuli (i.e., stimuli with 2 varying features). One set of subjects were trained to classify the stimuli into categories. A separate set of subjects received no training. Then, subjects were shown pairs of stimuli from the original training set, and asked to decide if the stimuli were the same or different (i.e., a perceptual discrimination task). Critically, Goldstone (1994b) found that subjects who went through the classification learning task had different patterns of perceptual sensitivity when making same-different judgments (specifically, greater sensitivity at category boundaries).

Goldstone's (1994b) study spurred a number of criticisms & follow-up studies exploring the nature of categorical perception within the TACL paradigm. One important issue regarding Goldstone's (1994b) original finding is whether or not subjects' perception of the stimuli actually change after learning. Rather than reflecting a shift in subjects' perceptual experience, Goldstone's (1994b) findings may have instead been due to subjects' strategic decision to respond with each exemplar's category label in mind (rather than the exemplar's "perceptual description"; Goldstone, Lippa, & Shiffrin, 2001). To counter this alternative account, Goldstone et al. (2001) trained subjects to classify categories of faces. Critically, subjects were then asked to make pairwise similarity judgments betweens exemplars from training *as well as* new generalization items with no category labels. Importantly, subjects who underwent classification training exhibited a perceptual sensitivity effect to not only the original training exemplars, but to the new items as well. Goldstone et al. (2001) took this as support for the hypothesis that classification learning does in fact alter subjects' perceptual experience of their

training environment, not just their response strategies.

A number of empirical demonstrations have further elucidated the nature of categorical perception in TACL experiments. Gureckis and Goldstone (2008) found that categorical perception effects occur not just at the boundaries of labeled categories, but also at the boundaries that divide unlabeled, within-category clusters of stimuli. This suggests that the phenomenon of categorical perception can occur as a result of unsupervised learning (i.e., category learning without explicit labels). In a similar vein, Folstein, Gauthier, and Palmeri (2010) found that unsupervised exposure to unlabeled stimuli impacts later category learning (particular when there were clear statistical patterns that described the features of the stimuli). Both of these studies suggest a mechanism through which statistical patterns (both continuous & discrete) shape humans' perception of their environment.

In a systematic investigation of categorical perception, Pothos and Reppa (2014) found evidence that categorical perception effects might be sensitive the particular cognitive demands required for learning different category structures. In their investigation, subjects were assigned to either an easy or difficult learning condition. Afterwards, subjects made pairwise similarity ratings of all possible stimulus pairings. Evidence of categorical perception was found only for subjects in the difficult learning condition. These results suggest that categorical perception serves a functional role in cognition. If categorical perception reflects some adaptation that makes category learning easier, then we wouldn't expect it to occur when cognitive demands are relatively low.

Goldstone's (1994b) and others' (Livingston et al., 1998; Gureckis & Goldstone,

2008; Folstein et al., 2010; Pothos & Reppa, 2014) work provide well-replicated evidence that category learning impacts similarity judgments after TACL experiments. However, there are a number of theoretical issues surrounding the concept of "similarity" in cognitive psychology. In the context of the TACL experiments described thus far, stimuli are defined by quantitatively describable features; that is, stimuli are represented in some "geometric" space Jäkel, Schölkopf, and Wichmann (2008). For example, a "dog" might be described in geometric space according to the continues values of its features (e.g., tail length, weight, furriness). This is also true of discrete or qualitative information as well. For example, the presence of a feature could be coded as 1 (present) or 0 (absent). Much of the foundational work in TACL has assumed a geometric representation of stimuli (Nosofsky, 1986; Kruschke, 1992; Kurtz, 2007; Shepard et al., 1961). There is an open debate about whether or not this stimulus representation meaningfully translates to the way in which stimuli are experienced by subjects in the lab (Jäkel et al., 2008; Tversky & Gati, 1982; Goldstone & Son, 2005).

Geometric stimulus representations make a number of assumptions about the nature of perceived similarity. That is, if stimuli are represented in the human mind as points in a geometric space, then subjects' similarity ratings should generally follow all the traditional axioms of geometry. For example, there is no distance between any point in geometric space and itself. If humans were to rate the dissimilarity between 2 copies of the same object, a geometric account predicts that their rating should be 0. Another assumption of geometric space is bi-directionality; that is, the distance between 2 objects should be the same regardless of the order of comparison (A = B : B = A). A human subject's similarity

rating of a cat and a dog should be the same regardless of whether they are asked "how similar is a dog to a cat" or "how similar is a cat to a dog". Additionally, the geometric account predicts that the pairwise similarity ratings between a set of 3 or more stimuli should conform to the properties of geometric shapes [4]. Critically, a number of studies have acquired similarity ratings that violate these assumptions (Tversky & Gati, 1982; Goldstone & Son, 2005; Podgorny & Garner, 1979; Holyoak & Gordon, 1983). Nevertheless, many of the more successful accounts of human category learning in the TACL paradigm rely on a geometric stimulus representation (Nosofsky, 1986; Kruschke, 1992; Love et al., 2004; Kurtz, 2015; Erickson & Kruschke, 1998; Nosofsky, Palmeri, & McKinley, 1994).

## 1.4 Connectionism and Categorical Perception

Many studies within the classification paradigm have found that categorical perception manifests itself behaviorally in a very particular way: objects from the same category are rated as "more similar" (*within-category compression*), while objects from different categories are rated as "less similar" (*between-category expansion*). If a learner's goal is to discriminate objects of different categories, then it seems intuitively useful for the learner to have category representations that shift their perception to make the task easier. This critical phenomenon within human cognition is elegantly embodied in a computational framework that's enjoyed both academic and industrial success: Connectionism.

Connectionism (McClelland, Rumelhart, Group, et al., 1987) is a biologically-

[4]For example, the distance between 3 points in space should follow the inherent properties of a triangle: the sum of the distances between point 1 & 2 and the distance between point 2 & 3 must always be greater than the distance between point 1 & 3 (Jäkel et al., 2008)

inspired framework that describes cognition as an emergent property of a dynamic network of interconnected processing units (or, "nodes"); these are often referred to as "artificial neural networks" (or, ANNs). While a great variety of ANN models exist, they generally share a number of core properties, namely:

- Information propagates through the network via artificial "synapses" (i.e., weighted connections between nodes),
- the signal that each node propagates (i.e., its "activation") is some aggregation of information from the environment or other nodes[5],
- over time, the weighted connections between nodes are updated to produce behaviors with greater fitness (i.e., learning).

The particular "function" of the network is often specified by the experimenter, and generally reflects some cognitive behavior that the experiment is trying to explain (e.g., classification, inference, decision-making, memory acquisition). Feed-forward ANNs are a popular style connectionist models where:

- networks are structured as a hierarchy of "layers",
- the first layer encodes information from the environment; the last layer produces some desired function[6],
- the weights of the network are optimized using gradient descent (a mathematical algorithm design to optimize any chain of parameterized mathematical operations).

[5]Often, a node's "activation" is realized through some nonlinear function (e.g., sigmoid function or a Gaussian distribution)

[6]information from the environment can be represented synthetically (e.g., symbolic features) or perceptually (e.g., raw images or sounds)

In its simplest form, a 2-layer feed-forward network (1 input layer, 1 output layer) is functionally equivalent to statistical regression. However, feed-forward networks can have as many or as few layers as needed to solve some task. Importantly, ANNs with many layers have (a) provided powerful theoretical accounts of human learning (Kruschke, 1992; Kurtz, 2007; Peterson, Abbott, & Griffiths, 2016), and (b) surpassed human-level performance on a number of simple, cognitive tasks (He, Zhang, Ren, & Sun, 2015). The flexibility of connectionist networks' architectures and functions has allowed them to computationally realize a variety of different theories of human categorization, include those from the reference point (Kruschke, 1992) & rule-based (Erickson & Kruschke, 1998) framework.

While connectionist networks have provided excellent fits to human learning and generalization within the TACL paradigm, they provide an interesting avenue for research on categorical perception as well. Categorical perception studies have often found that category learning influences perception in a way that intuitively seems to make the learning task easier (Goldstone, 1994b; Goldstone et al., 2001). Within the connectionist framework, an encountered stimulus is "represented" as the activation of 1 or many nodes in a network. This representation changes during learning in order to improve the networks ability to solve some task, which is arguably an example of categorical perception (Tijsseling & Harnad, 1997).

While connectionist models demonstrate their own form of categorical perception, it is an open question whether the "representations" in connectionist networks are a meaningful analogue to the abstract notion of stimulus representations in the human mind. However, a number of recent studies have demonstrated that the representations learned by connectionist models (particularly in the domain

of vision) are predictive of various aspects of human behavior as well as human physiology (Peterson et al., 2016; Long, Yu, & Konkle, 2018; Lake, Zaremba, Fergus, & Gureckis, 2015). In order to make such a demonstration, researchers need some analysis to assess "representational similarity" across models and humans. Recent work has made a number of steps towards this goal (Kriegeskorte, Mur, & Bandettini, 2008).

Traditional feed-forward connectionist networks map stimuli to responses by transforming featural information through 1 or many layers of the network. Typically, each layer of the network produces a unique activation for each stimulus. Laakso and Cottrell (2000) proposed that researchers could assess the stimulus representations of feed-forward connectionist networks by computing differences in stimulus activations at a given layer (see figure 1.2). This type of analysis is ideal in that it is model-invariant, allowing researchers to compare the hypothetical "representation space" of any animal or computational model that produces some response to a stimulus or event in the environment.

Khaligh-Razavi and Kriegeskorte (2014) found that supervised, convolutional neural networks (or CNNs, a feed-forward, connectionist network designed for categorizing perceptual stimuli) were very predictive of physiological responses to stimuli in humans and non-human primates (relative to other leading models of animal vision). The correlation between BOLD responses or single-neuron responses for each stimulus was used as a metric for the "representational space" of humans or Macaque monkeys (respectively). Critically, the representational space of various layers of CNNs were very predictive of the representational space observed in the physiological responses of humans and Macaques. More specif-

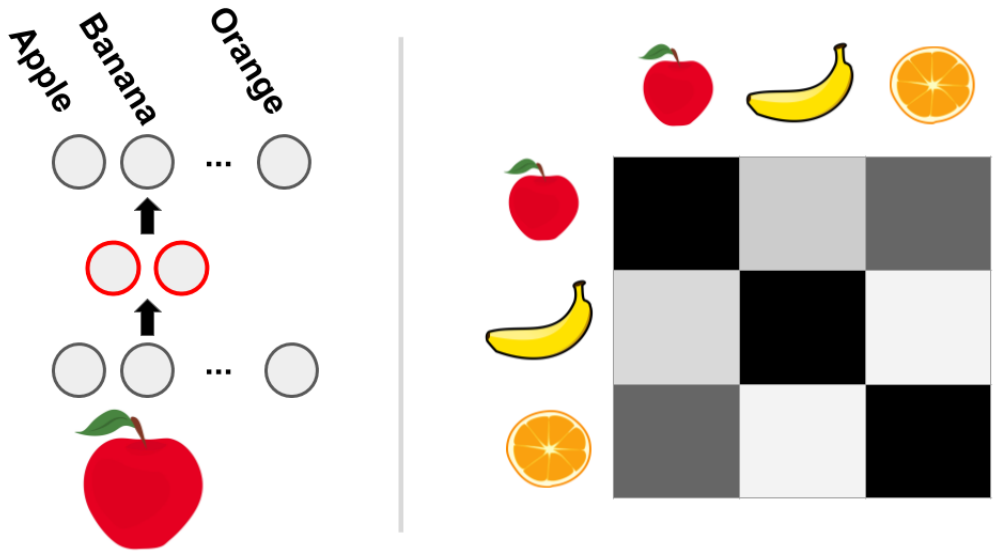


Figure 1.2: Each stimulus produces a unique activation at the hidden layer of the network. The differences between all unique pairs can be plotted as a *representational matrix*, where each square represents the similarity of activations between 2 stimuli.

ically, the earlier layers of CNNs (which are thought to detect low level image statistics; Lee, Grosse, Ranganath, & Ng, 2009) were more predictive of the earlier layers of the visual pathway (V1, V2), while the later layers of CNNs (which are complex aggregations of patterns from the lower layers) were more predictive of the later stages of the visual pathway (V7, IT). Long et al. (2018) found a similar result in human BOLD responses in the ventral stream, using both intact images as well as images that were perceptually transformed to be unrecognizable by human subjects[7]. This suggests that both humans and CNNs are sensitive to category-specific texture information of perceptual objects[8].

Peterson et al. (2016) demonstrated that the representational space of deep connectionist networks are also predictive of human behavior. In their work, Pe-

[7]Interesting, Long et al. (2018) found that behavioral similarity ratings predicted BOLD responses as well.

[8]Importantly, CNNs may lack sensitivity to information beyond surface textures, which may limit their viability as a model of human cognition (Baker, Lu, Erlikhman, & Kellman, 2018)

terson et al. (2016) gathered pairwise similarity ratings of 120 images of animals. They found that the representational space of modified, pre-trained CNNs accurately predicted the pairwise similarity judgments of human subjects. In similar work, Lake et al. (2015) found that CNNs accurately predicted category-typicality ratings of natural objects as well. While this doesn't directly suggest a mechanistic overlap between humans and connectionist networks, it does provide insight to the statistical properties of the environment that humans might be sensitive to.

#### Relevance to Present Work

In the TACL paradigm, the goal of the learner is to accurately classify the category of a stimulus. Evidence from similarity ratings suggest that classification learning impacts the way humans learners "perceive" experienced stimuli[9]. In connectionist models, classification learning impacts the way the model "represents" experienced stimuli. If stimulus representations in connectionist networks can predict human similarity judgments (Peterson et al., 2016), then connectionist models could provide insight into how human category representations shift during TACL experiments.

## 1.5 Limitations of TACL and Alternatives to Classification Learning

The TACL paradigm has many nuances that make it particularly useful for investigating human category learning and the computational models that try to explain it. However, the standard TACL paradigm only encompasses a narrow

[9]or at least, how they report to perceive them

domain through which categories are learned (Markman & Ross, 2003). In classification learning, the objective of the learner is to accurately classify stimulus labels (Kurtz, 2015; Nosofsky, Gluck, et al., 1994). Importantly, research has shown that this particular learning goal biases the type of information subjects pay attention to (Markman & Ross, 2003; Levering & Kurtz, 2015; Chin-Parker & Ross, 2004).

In a standard TACL experiment, subjects are shown a stimulus (or, a set of features), and asked to guess its category label. It is therefor advantageous for subjects to focus their attention towards features that are *diagnostic* of category membership (as opposed to features that are *prototypical*[10] of category membership). In an alternative task, Chin-Parker and Ross (2004) modified the standard TACL design and presented subjects with an already classified stimulus with a missing feature; subjects were then asked to guess the missing feature's value. Here, the subjects goal isn't to classify objects into categories, but rather to infer their category-specific properties (i.e., inference learning). Subjects who underwent either a classification or inference training phase were then asked to make category judgments about one or two features (rather than about entire stimulus). Importantly, results from the feature classification test demonstrated that inference learners were more sensitive to *prototypical* features while classification learners were more sensitive to *diagnostic* features.

The inference learning task utilized by Chin-Parker and Ross (2004) highlights a condition in which subjects are more sensitive to *prototypical* properties of a category. Levering and Kurtz (2015) also demonstrated that diagnostic bias can

[10]Prototypical information can be described as featural properties common to many exemplars of a category

be reduced when subjects are just told to *observe* labeled stimuli (instead of making classification decisions). Levering and Kurtz (2015) found that observation learners were *more* sensitive to *generative* properties of categories (namely, central tendency of feature values & correlations between features). Importantly, however, learners in the observation condition were still generally biased towards learning diagnostic properties of features.

Endorsement learning is a learning mode where subjects decide whether or not a stimulus belongs to acandidate category label (Levering & Kurtz, 2015; Kattner, Cox, & Green, 2016). Unlike classification, endorsement learning does not explicitly ask subjects to choose between a mutually exclusive list of potential categories. Rather, subjects decide whether or not to endorse a candidate label as a correct descriptor of a stimulus. Kattner et al. (2016) trained subjects on categories made up of 2 dimensional stimuli clustered according to 1 diagnostic feature. Subjects learned the categories via either (a) a classification learning task (where categories could be differentiated according to a single dimensional rule), or (b) an endorsement learning task (where categories were described according to a narrow range along a single dimension). Critically, Kattner et al. (2016) found that endorsement learners were less likely to form diagnostic, rule-based representations of learned categories. Given that a number of studies have demonstrated qualitative empirical differences between various modes through which categories are learned (Markman & Ross, 2003; Levering & Kurtz, 2015; Chin-Parker & Ross, 2004; Kattner et al., 2016), it is an open question what impact different learning modes have on the manifestation of categorical perception after category learning.

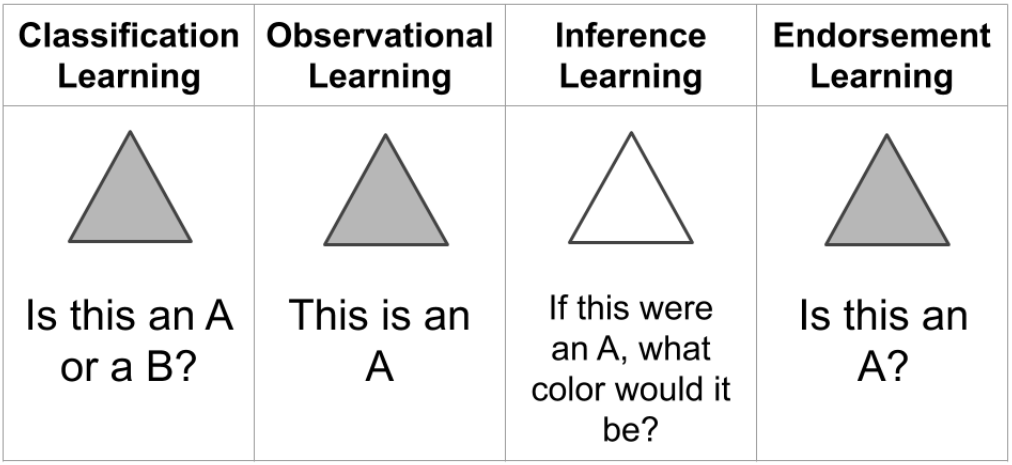


Figure 1.3: Comparison of 4 different types of learning modes.

## 1.6 The Present Work

The present work aims to shed light on 2 primary questions of interest. First, it is of theoretical interest whether learning mode impacts the nature of categorical perception effects during artificial category learning tasks. The categorical perception effects typically reported in the literature suggest that classification learning leads to an increase in within-category similarity ratings and a decrease in between-category similarity ratings[11]. Intuitively, if the learner's goal is to tell things apart, it would be beneficial to perceive objects from different categories as being more dissimilar than they were before learning. Research has shown that alternative learning objectives (e.g., inference, observation, endorsement) lead to different behavioral effects after learning (Levering & Kurtz, 2015; Chin-Parker & Ross, 2004; Kattner et al., 2016). It is currently unknown whether different learning objectives will lead to qualitatively different patterns of similarity ratings

[11] i.e., alike things become more alike; unalike things become more unalike

as well. To this end, the present work contrasts classification learning with endorsement learning. In classification learning, subjects decide the correct category label of a provided example. In endorsement learning, subjects decide whether a provided category label accurately describes a provided example. Prior work has suggested that endorsement learning leads to more generative generalization profiles; it is therefor predicted that endorsement learning will lead to more generative similarity ratings as well (ie, less between-category dissimilarity).

The second primary question of interest is whether leading connectionist accounts of human categorization can account for task-specific, categorical perception effects. Typically, computational models of categorization are evaluated based on their success at predicting ease of learning (i.e., how long it takes subjects to correctly classify objects after a number of training blocks; Kurtz, 2015; Nosofsky, Gluck, et al., 1994) and generalization (i.e., the category label subjects assign to objects they've never seen before; Erickson & Kruschke, 1998; Nosofsky & Johansen, 2000). Given that this has left the field with a number of successful (yet fundamentally distinct) theoretical accounts (Wills & Pothos, 2012; Kurtz, 2015), the present work provides an additional challenge to computational models: a successful account of the present results should be able to predict not only ease of acquisition & generalization, but also the similarity ratings of human learners.

# 2 Behavioral Experiment

## 2.1 Methods

### 2.1.1 Participants

A total of 284 Binghamton undergraduate students were randomly assigned to 1 of 9 conditions of the experiment (3 learning modes & 3 category structures). Subjects received course credit towards a degree requirement as compensation for participation.

| | *Category Structure* | | | |
|---|---|---|---|---|
| | **0-Share** | **1-Share** | **2-Share** | |
| **Classification** | 34 | 33 | 35 | 102 |
| **Endorsement** | 31 | 28 | 32 | 91 |
| **Control** | 30 | 29 | 32 | 91 |
| *Learning Mode* | 95 | 90 | 99 | |

Table 2.1: Distribution of subjects across conditions

### 2.1.2 Design

Subjects were assigned to 1 of 3 learning groups: a *Classification* group, an *Endorsement* group, and a *Control* group (no learning). In the Classification group, subjects engaged in 7 blocks of supervised classification learning, followed by 1 block of unsupervised classification testing. After, subjects provided similarity ratings to pairs of stimuli from the training set. In the Endorsement group, subjects engaged in 7 blocks of supervised endorsement learning, followed by 1

block of unsupervised endorsement testing; this was also followed by a similarity rating phase (identical to the Classification group). In the Control group, subjects only provided similarity ratings to pairs of stimuli (no learning or testing preceded this phase of the experiment). There were two types of similarity ratings for all groups: (1) within-category pairs (two stimuli from the same category) and (2) between-category pairs (two stimuli from different categories). Importantly, prior work on categorical perception suggests that classification learning induces higher within-category similarity ratings and lower between-category similarity ratings relative to control (Goldstone, 1994b; Goldstone et al., 2001; Pothos & Reppa, 2014).

Stimuli were grouped into 3 categories described by 2 relevant features. Within each learning group, subjects could be assigned to 1 of 3 category structures, labeled: 0-Share, 1-Share, or 2-Share (see figure 2.1). In the 0-Share structure, none of the categories share overlapping feature values. In the 1-Share structure, 1 pair of categories overlap on 1 feature. In the 2-Share structure, all pairs of categories share overlapping values on each relevant feature. These particular category structures can be solved with different attentional solutions; that is, both the 0-Share and 1-Share structures can be solved by only attending to 1 relevant feature.

### 2.1.3 Materials

The experiments were conducted on Windows PCs using the program PsychoPy (Peirce et al., 2019). The training stimuli consisted of 21 squares that varied in their size (ranging from 2.5cm - 7.1cm) and shading (ranging from 25 -

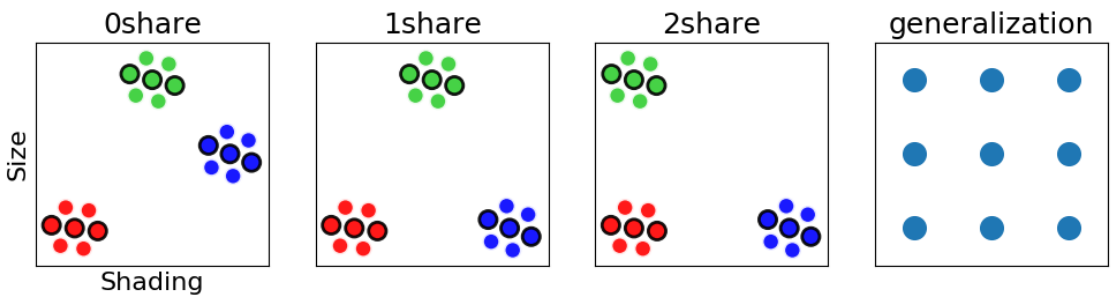


Figure 2.1: Category structures used in the experiment. Each point is a stimulus represented in a 2 dimensional feature space. Points that are outlined were used for the similarity ratings phase. Generalization stimuli were not seen during training.

230 greyscale[1]) and were evenly divided among 3 categories (labeled Alpha, Beta, or Gamma). The generalization stimuli evenly spanned to the space of possible feature values not captured by the training stimuli (figure 2.1). The generalization stimuli provided an indication of how subjects generalized their beliefs about the categories across the space of possible stimuli.

### 2.1.4 Procedure

In the Classification and Endorsement groups, subjects began the first supervised training phase after reading a prompt describing the instructions for the study. In the Classification group, subjects were presented with a stimulus, and asked to guess which of 3 provided categories they believed the stimulus belonged to (figure 1.3). After, subjects were told whether their guess was correct or incorrect. In the Endorsement group, subjects were provided with both a stimulus and 1 possible category label; their task was to decided whether or not the stimulus belonged to the provided label (figure 1.3). They were also given corrective feedback after their guess. Each of the 21 stimuli was shown once per block. This resulted in a total of 7 * 21 = 147 trials.

[1]Greyscale values range from 0 (black) to 255 (white)

After training, each group underwent 1 block of unsupervised testing that included the original 21 training stimuli, as well as 6 generalization stimuli not seen during training (for a total of 27 trials). In the test phase, no corrective feedback was provided after subjects made a classification or endorsement decision. Importantly, the task during the test phase was the same as the task subjects experienced during training.

All learning groups ended the experiment by providing similarity ratings for pairs of stimuli from the training phase. Because a set of 21 stimuli results in a large number of possible stimulus pairs (210 unique pairs), only a subset of the 9 stimuli were included in the similarity rating phase (figure 2.1): 3 from each category, resulting in a total of 36 pairwise ratings from each participant. Notably, subjects did not provide ratings between a stimulus and itself.

## 2.2 Results & Discussion

### 2.2.1 Learning Difficulty

Various theoretical models of human categorization make a priori predictions about the relative difficulty of learning different category structures (Kruschke, 1992; Kurtz, 2007; Love et al., 2004; Anderson, 1991). By the final block of learning in all conditions, accuracy converged to near perfect performance (figure 2.2). To assess differences in learning acquisition across conditions, each subject's learning data was quantified according to how quickly the subject reached perfect accuracy (hereby referred to as *first perfect block*). One shortcoming of this coding scheme is that it is difficult to describe learners who never achieved mastery (perfect accuracy by the final block of training). Two potential ways of remedying

this limitation are to (1) remove learners who didn't achieve some experimenter defined threshold of mastery[2], or (2) give learners who didn't acquire mastery an arbitrary score. For this analysis, the latter option was chosen to avoid removing meaningful data about the experiment (i.e., what conditions resulted in learning failures); subjects who didn't achieve mastery were given a *first perfect block* score of 8 (1 more than the number of blocks in the learning phase of the experiment). [3]

The *first perfect block* scores suggest that classification learners reached mastery sooner ($M$=2.647, $SD$=1.199) than endorsement learners ($M$=4.725, $SD$=2.044) when averaging across category structure. There appeared to be little difference in the rate mastery between the 0-Share condition ($M$=3.862, $SD$=1.911), the 1-Share condition ($M$=3.443, $SD$=2.013), and the 2-Share condition ($M$=3.567, $SD$=1.932) when averaging across learning mode. A two-way ANOVA revealed a significant main effect of learning mode, $F(1,187)=75.9$, $p<.001$. However, there was no significant main effect of category structure, $F(2,187)=0.8$, $p>.05$, and no significant interaction, $F(2,187)=1.2$, $p>.05$.

In summary, the results suggest that endorsement learning takes more exposures to master than classification learning. This could be because endorsement learning encourages subjects to learn *generative* properties of the categories (which may not be directly diagnostic of category membership). Alternatively, the need for more exposure in the endorsement condition might reflect ambiguity in the feedback given on each trial. In classification learning, subjects are informed of

[2]Only 8.3% of learners failed to achieve mastery; all of them were in the endorsement condition

[3]Both methods of describing subjects who failed to learn produced qualitatively similar results.

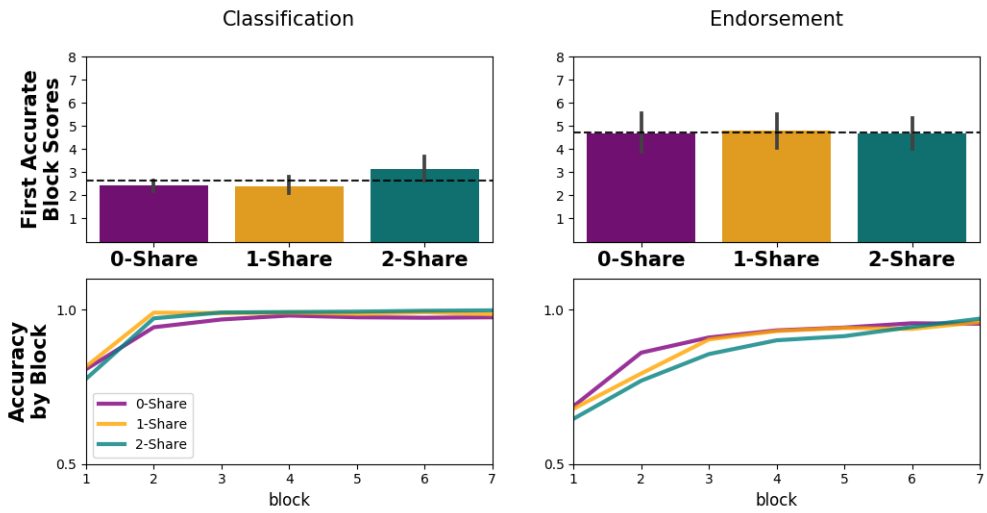


Figure 2.2: **Top Row:** Averaged block in which subjects achieve perfect accuracy. The dashed horizontal line indicates the mean score (averaged across category structure). Vertical bars indicate 95% confidence intervals. **Bottom Row:** Learning curves for each category structure plotted as the mean accuracy and each block of training.

the correct category membership of each stimulus on each trial. In endorsement learning, subjects are only given complete feedback when the subjects correctly identify that the presented label matches the presented stimulus; on all other trials, the feedback given to subjects is partially ambiguous (figure 2.3).

| | **Yes Response** | **No Response** |
|---|---|---|
| **Correct Label** | Complete Feedback | Complete Feedback |
| **Incorrect Label** | Incomplete Feedback | Incomplete Feedback |

Figure 2.3: All possible response scenarios for endorsement learning trials. Subjects only get complete feedback when the presented label matches the presented stimulus.

It was expected that structures with more opportunities for a attentional solution would be easier to learn. However, the results suggest that 3 category structures were learned at roughly the same rate. This might suggest that the

number of attentional solutions does not impact the average rate at which the category structures are mastered. An alternative possibility is that the category structures were relatively easy to learn (only 8.29% of subjects failed to reach mastery at the end of training); this ceiling effect may have suppressed any observable differences in the rate at which the category structures were learned.

### 2.2.2 Similarity Ratings

Recall that categorical perception affects are defined as a change in similarity ratings between objects as a function of category learning. Typically, this manifests itself as an *increase* in within-category similarity ratings (compression), and a *decrease* in between-category similarity ratings (expansion) relative to control. One primary objective of the present work was to examine the impact of learning mode on categorical perception effects. Across all conditions, each subject provided a similarity rating for all possible combinations of stimuli[4] (figure 2.5 highlights the mean similarity ratings for each stimulus pair in each condition).

Importantly, the present work partially replicates the typically observed pattern of similarity ratings as a function of learning. Subjects pairwise similarity ratings were broken down into 2 types: between-category ratings & within category ratings. For the between-category similarity ratings, a two-way ANOVA revealed a statistically significant main effect of learning mode, $F(2,257)=21.688$, $p<.001$, a significant main effect of category structure $F(2,257)=4.610$, $p<.011$, and no significant interaction $F(5,257)=1.728$, $p>.05$. The main effect of category structure is expected given that the category structures differed in their

[4]Subjects were not asked to provide similarity ratings between a stimulus and itself.

inherent between-category distances and will not be explored further. Regarding the main effect of learning mode, post hoc t-tests revealed that subjects in the control condition rated between-category stimulus pairs as more similar ($M$=53.9, $SD$=18.6) than subjects in the classification condition ($M$=42.3, $SD$=15.1), $t(182)$=5.204, $p_{bonf}$<.001, and the endorsement condition ($M$=40.043, $SD$=11.891), $t(171)$=6.067, $p_{bonf}$<.001; this replicates the *between-category expansion* effect typically observed in the categorical perception literature (Goldstone, 1994b). Importantly, between-category ratings in the classification condition did not differ significantly from the endorsement condition, $t(173)$=1.037, $p_{bonf}$>.05. For the within-category similarity ratings, a two-way ANOVA revealed no significant main effect of category structure, $F(2,257)$=.076, $p$>.05, no main effect of learning mode, $F(2,257)$=.661, $p$>.05, and a significant interaction[5] $F(4,257)$=3.324, $p$<.011; this fails to replicate the *within-category compression* effect typically observed in the literature.

The main effect of learning mode on between-category similarity ratings suggests that the category learning tasks (both classification and endorsement) led subjects to rate between-category stimulus pairs as *less similar* than subjects who did not undergo any categorization training (control condition). Contrary to the hypothesis, endorsement learning did in fact produce a between-category expansion effect, suggesting that subjects may have learned discriminative properties of the category structures. Critically, the was no evidence of *within-category compression* effects in either learning mode. This failure to replicate may reflect a ceiling effect; this seems plausible given that the maximum possible similarity

[5]Because none of the post-hoc tests were significant, this interaction is not explored further.

rating (100) fell within 2 standard deviations from the mean for all 3 conditions. Another related possibility is that task was too easy to induce any adaptive change in category representations.

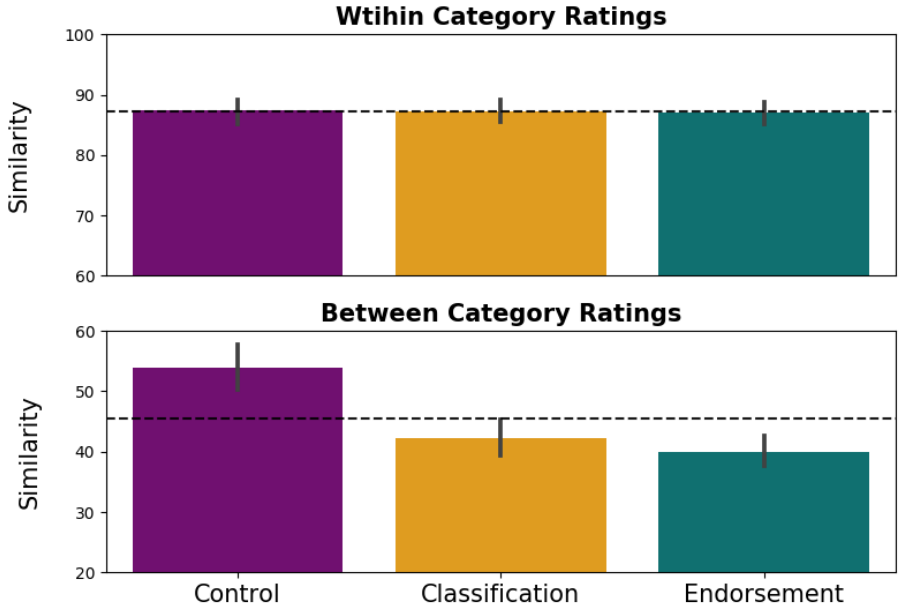


Figure 2.4: Mean similarity ratings for each learning mode. Horizontal dashed line represents the mean across learning modes; vertical bars represent 95% confidence intervals. An effect of learning mode was only found for *between-category* similarity ratings, indicating that ratings in the endorsement and classification differed from control.

### 2.2.3 Post-Hoc Analyses

#### Generalization Gradients

In the unsupervised test phase of the experiment, subjects were asked to categorize the original stimuli from the training phase as well as new unseen items that spanned the remaining feature space. This particular analysis helps elucidate the knowledge that subjects learned about the categories. The patterns of generalization can be analyzed by aggregating the generalization responses for each stimulus across subjects (figure 2.6 provides a visualization of subjects generalization profiles for each condition). This particular analysis suggests a correspondence

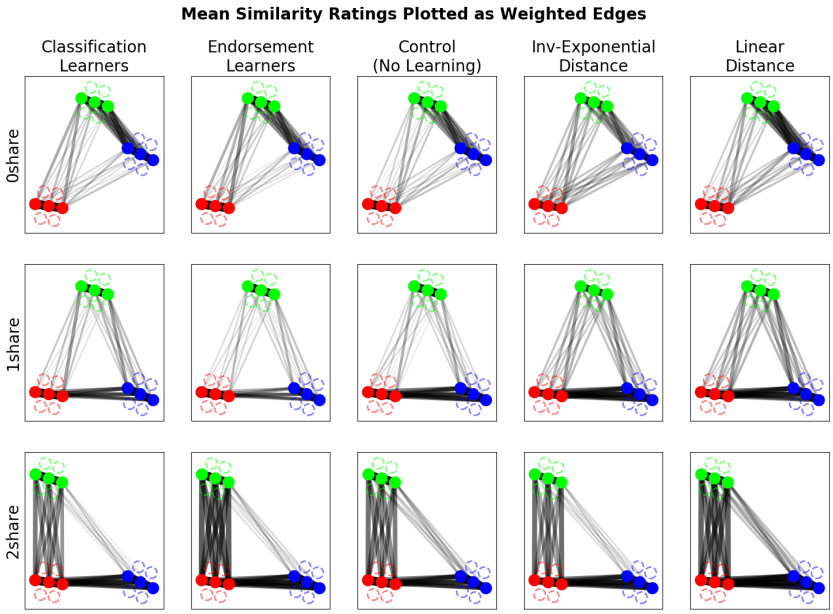


Figure 2.5: Graphical representation of similarity ratings for each condition. Each node represents a stimulus, each edge represents the mean similarity rating for 2 stimuli. An Inverse-exponential distance function (as well as a linear distance function) provides a fairly robust account of the similarity ratings for each pair.

between the generalization behavior across learning modes, and also suggests that subjects may have used attentional rules to solve the problem; this warrants a more detailed analysis of individual subjects.

While the aggregate generalization profiles appear to be very similar across learning mode for the 0-Share and 1-Share condition, they differ critically in the 2-Share condition, particularly in regards to the generalization stimulus equidistant to all categories. In the classification condition, this stimulus was typically classified as if subjects were using an attentional boundary that isolated the categories with higher values on either dimension (importantly, 10 subjects in the 2-Share condition classified the stimulus in this way). This was not the case for the endorsement condition, which may suggest that endorsement learners were less likely to utilize multi-faceted attentional solutions to solve the problem.

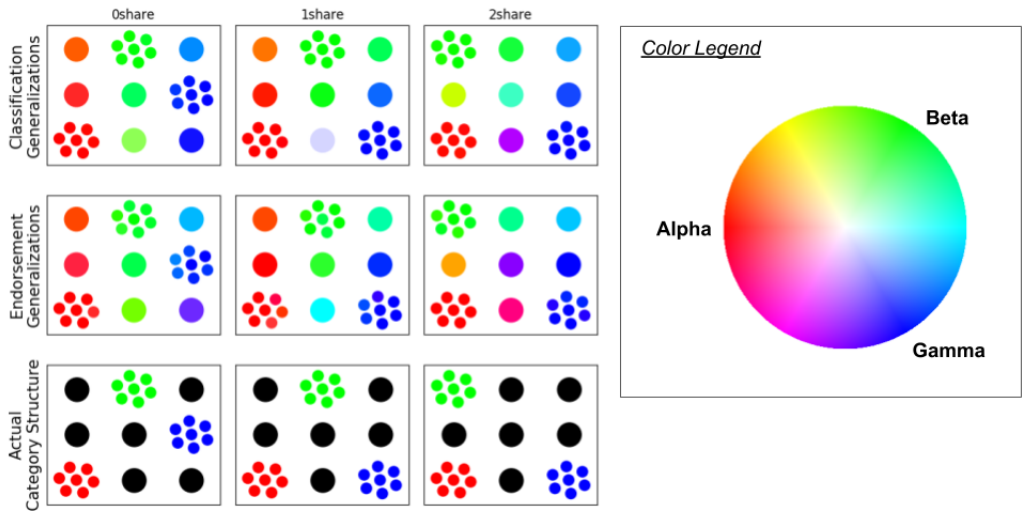


Figure 2.6: Aggregated generalization profiles for each condition. Response probabilities were converted to RGB values for each category (red for the "Alpha" category, green for "Beta", and blue for "Gamma"; white indicates that a stimulus was equally likely to be assign to either of the 3 categories.

### Attentional Learners

The 3 category structures in this experiment varied in the number of available attentional solutions. Consequently, it is of interest whether subjects utilized these attentional solutions at all. The aggregated generalization data suggests learners were indeed utilizing attentional solutions. This warrants a more fine-grained, subject level analysis that queries each subject's classification decision for all of the test items. This information is available for the classification condition, since each classification learner provided a category label for each stimulus. However, this information is mostly unavailable for the endorsement condition at the subject level (since endorsement learners only provide Yes/No decisions). When looking at the attentional profiles of subjects in the classification condition, it is clear that many subjects utilized attentional solutions when they were available.

In the 0-share condition, subjects could perfectly solve the task by only pay-

ing attention to either of the 2 relevant features of the stimuli. In the 0-share condition, subjects were identified as *attentional learners* if their generalization profile perfectly matched either of the 2 possible attentional templates (single feature solution; figure 2.7). 15 subjects matched this definition (figure 2.7). In the 1share condition, only 1 of the features is need to solve the classification task. 7 subjects utilized a purely attentional solution for the 1-share condition. In the 2-share condition, subjects cannot solve the task by only attending to a single feature. However, 10 subjects' generalization profiles suggest the generalization of a category along a complete unidimensional boundary (figure 2.7). While many subjects' generalization profiles indicate a learned attentional bias, this did not seem to impact those subject's overall pattern similarity ratings[6].

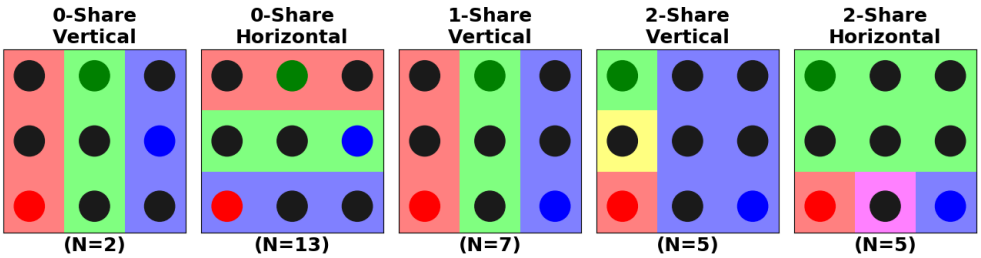


Figure 2.7: Subjects were classified as an attentional learner if their generalization profiles matched the expected profile of a given template. Colored dots represent categories; black dots represent generalization items.

### Change as a function of Feature Distance

The results of the current work replicated a typical finding in the categorical perception literature: objects from the same category are rated as "more similar", and objects from different categories are perceived as "less similar". This

[6]Note, however, that this particular analysis aggregates over all between-category & within-category stimulus pairings. This might mask structural patterns in the similarity data.

phenomenon might be an adaptive strategy towards solving a classification task: if things from different categories are perceived as being less similar as than they inherently are, it might be easier to tell them apart. This might be especially useful when the distance between category boundaries is very small, and less useful when categories are already easier to distinguish. If that were the case, we would expect that the magnitude of categorical perception effects should be larger when between-category stimulus-pairs are very similar[7].

To test this prediction, the pairwise similarity ratings were first averaged across subjects. The *magnitude* of categorical perception effect was then defined as a difference score between the similarity ratings for the control condition and the similarity ratings for each learning condition (classification, endorsement). This was created for each unique stimulus pairing. The inherent similarity between two stimuli was defined as the absolute difference between the feature vector of each stimulus. For between-category similarity ratings, a pearson correlation revealed that the difference scores between the control and learning conditions were negatively correlated with the raw featural distances of the stimuli, $r(160)$ = -.307, $p$<.001. For the within-category similarity ratings, a pearson correlation revealed no significant correlation between difference scores and featural distance, r(52)=.1, $p$>.05[8].

[7]The opposite effect should occur for within-category stimulus pairs

[8]A limitation in this analysis is the apparent ceiling effect observed in within-category similarity ratings, as well as the small number of within-category stimulus pairs to compare with.

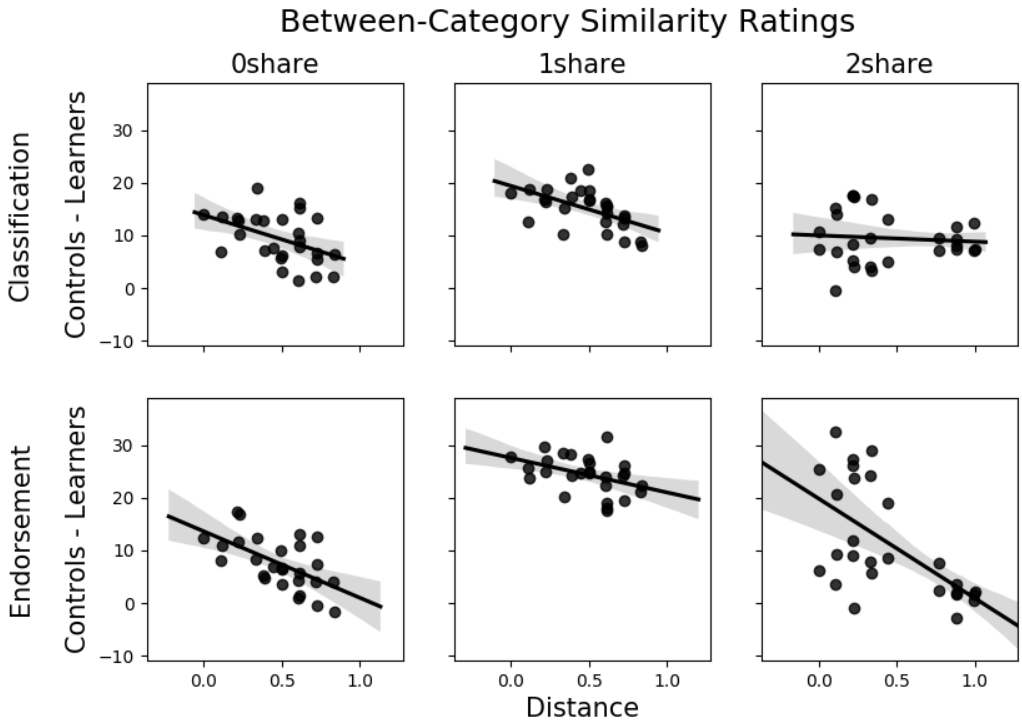


Figure 2.8: Between-category expansion effects plotted as a function of spatial distance between stimulus pairs for between-category similarity ratings. The y-axis shows the difference in similarity ratings between the learning and control conditions for each group (i.e., between-category expansion). As the distance between 2 stimuli increases, the magnitude of between-category expansion decreases

### Summary

In summary, the present work replicates the between-category expansion effect typically observed in the categorical perception literature, in addition to extending this finding to endorsement learning[9]. The *first perfect block* analysis suggests that endorsement learning takes more exposure to attain mastery than classification learning. Regarding generalization, while a number of learners in the classification phase solved the task using a unidimensional solution, this strategy didn't appear to have an observable impact on their aggregated similarity ratings. Lastly, differences in between-category similarity ratings between the learning conditions

[9]contrary to what was originally hypothesized

& the control condition were mediated by the featural distance of stimulus pairs. To test the psychological plausibility of competing computational theories, candidate models will be assessed according to how well they predict the similarity ratings of this experiment. However, a complete account should be able to predict the ancillary findings as well.

# 3 Computational Experiments

The categorization literature is now filled with a number of different models that provide relatively good predictions of human behavior during classification learning on benchmark problems (Wills & Pothos, 2012; Kurtz, 2015). One factor might be that human categorization is a multi-faceted process that requires different frameworks to fully describe[1]. Another factor might be that many of the most successful models of human categorization are flexible enough to produce any particular behavior (i.e, they are no longer falsifiable; Wills & Pothos, 2012). One way to resolve this lack of falsifiability is to require models to provide a more comprehensive, multi-faceted account of empirical phenomena. The present work explores whether 2 existing theoretical models in the literature (Kruschke, 1992; Kurtz, 2007) can adequately predict both (a) human learning acquisition, and (b) changes in similarity ratings during a category learning task. The present work also introduces a category-learning model that utilizes a common representation of stimuli across different tasks (specifically, classification & inference).

[1]in part suggested by dual-systems theories (Erickson & Kruschke, 1998)

## 3.1 Candidate Models

### 3.1.1 Attention Learning COVEring map

Exemplar models posit that classification decisions are made by generalizing from stored instances of previously categorized stimuli in memory (Nosofsky, 1986; Medin & Schaffer, 1978). A connectionist instantiation of exemplar models, ALCOVE (Kruschke, 1992), has been widely successful at predicting a number of behavioral phenomena in the categorization literature (Kruschke, 1992; Nosofsky, Gluck, et al., 1994), particularly in regard to classification learning. The next section will briefly describe how ALCOVE operates.

ALCOVE can be considered a non-parametric way of building a probability density function within a continuous stimulus space (Jäkel et al., 2008; Rosseel, 2002); importantly, dimensions of this stimulus space are weighted to bias task-relevant features over irrelevant features. During classification learning, ALCOVE first represents a stimulus according to its similarity to a set of reference points that lie in the stimulus space (these reference points are represented as radial basis functions centered at the coordinates of the stimuli seen during training). The reference points are analogous to hidden nodes of the hidden layer in a feed-forward connectionist network. The activation of ALCOVE's hidden layer of reference points $a_h$ for each stimulus $X$ is given by the equation:

$$a_h = e^{-c((|X-h|\cdot W_\alpha^T)^r)^{1/r}} \tag{3.1}$$

Where $W_\alpha$ is a set of attentional weights to each stimulus feature, $h$ is a vector of feature values representing the hidden nodes, $r$ reflects the geometry of the

distance metric[2], and $c$ is a specificity parameter that determines the size of the radial basis function. A classification decision for each stimulus is decided by the summed activations of each reference point weighted by the learned associations between reference points and known categories. The probability that a stimulus will be categorized as a member of category $K$ is given by the equation:

$$P(K) = \frac{e^{\phi a_h \cdot W_K}}{\sum\limits_{k=1} e^{\phi a_h \cdot W_k}} \tag{3.2}$$

Where $w_k$ represents the learned association weights for a given category and $\phi$ is a response mapping parameter that determines the sharpness of classification probabilities across categories. Importantly, the attention and association weights of ALCOVE are update trial-by-trial using gradient descent to minimize the error between the true category label of a stimulus and ALCOVE's classification response.

Despite their relative empirical success at explaining human classification learning, exemplar models have the problem of being able to adequately learn any arbitrary probability function. This makes it difficult for exemplar models to predict the ease through which categories are learned (Rosseel, 2002). ALCOVE provides one solution to this limitation by implying that subjects need experience to learn which features and exemplars are most relevant for classification decisions. In additional limitation of exemplar models is the requirement to remember all exposures to exemplars of a given category. Interestingly, however, reduced versions of exemplar models that *learn* the coordinates of (arbitrary) reference points (e.g., *mixture models of categorization*) have provided behavioral predictions of hu-

[2]e.g., city-block ($r = 1$) or euclidean ($r = 2$)

man learning on-par with traditional exemplar models on some problems (Rosseel, 2002).

### Categorical Perception in ALCOVE

ALCOVE's hidden layer represents stimuli according to their attention-weighted similarity to stored reference points in memory. The attention-weighted similarity in ALCOVE provides an intuitive explanation of categorical perception effects. When biasing attention towards features that are diagnostic of category membership (and suppressing attention towards features that are not diagnostic), ALCOVE inflates the distance between categories and reduces the distance within categories (see figure 3.1). This makes ALCOVE's hidden layer representation and ideal predictor of similarity change in the context of the present study.

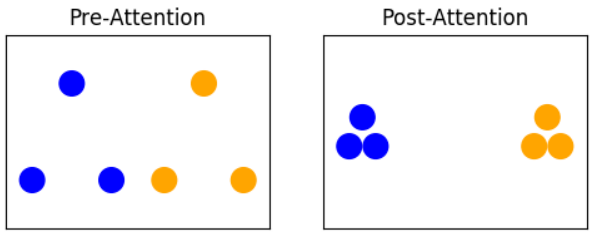


Figure 3.1: Impact of ALCOVE's learned attention weights on between-category and within-category similarity. Stimuli are compressed along nondiagnostic dimensions, and expanded along diagnostic dimensions.

### 3.1.2 DIVergent Autoencoder

Exemplar models posit that classification learning is a process of associating category labels with remembered regions of an attention-mediated stimulus space; i.e, the probability of the category label given some stimulus (data):

$$p(C_k|D)$$

Another approach to classification learning is to instead try to infer the properties of the data (stimuli) given some category (Levering & Kurtz, 2015):

$$p(D|C_k)$$

This generative approach to classification learning is embodied by the DIVergent Autoencoder (DIVA; Kurtz, 2007) model of human categorization. DIVA is a feed-forward connectionist network that learns statistical descriptions of experienced stimuli; classification decisions are made according to which category-specific description of the stimulus best reflects the models current experience. These category-specific descriptions typically contain information about statistical relationships between the features of stimuli. The next section will briefly describe how DIVA operates.

An autoencoder is a feed-forward connectionist model that can encode information in a nonlinear representational space to make statistically optimal inferences about data (Kramer, 1991). This nonlinear representational space emerges from the combined activity of the autoencoder's "hidden layer". During learning, autoencoders first represent a given stimulus by filtering that stimulus through a set of weights connecting features of the stimulus to nodes in the hidden layer; the "activation" of the hidden layer is given by the equation:

$$a_h = \sigma(D \cdot W_1 + b_1) \tag{3.3}$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{3.4}$$

Where $D$ is a vector describing the properties of the stimulus (i.e., features), $W_1$ and $b_1$ are a set of learned weights connecting stimulus features to hidden nodes, and $\sigma$ is a nonlinear, logistic *activation function*. Next, the hidden activation is fed through an additional layer of weights optimized to accurately reconstruct an experienced stimulus from its hidden representation; the output of an autoencoder is given by the equation:

$$a_o = a_h \cdot W_2 + b_2 \tag{3.5}$$

Where $W_2$ and $b_2$ are reconstructive weights connecting the hidden layer to the output layer. The weights of an autoencoder are optimized using gradient descent to minimize the reconstructive error between the model's output activation[3] and its current experience (the data).

DIVA is a variant of autoencoders that learns a distinct set of reconstructive weights $W_k$ for each category $k$ within some learning domain. During learning, each set of reconstructive weights (or, *category channel*) is only exposed to stimuli that belong to the category it was assigned to. A consequence of this procedure is that category-specific weights should work best when reconstructing stimuli from their specific category. Naturally, DIVA makes a classification decision by determining which category channel provides the most accurate description of the stimulus (i.e., the category channel with the lowest reconstructive error); this is given by the equation:

[3]Because the model's output activation is a new representation of the original stimulus, it will have the same number of dimensions as the original stimulus.

$$P(K) = \frac{1/SSE_k}{\sum 1/SSE_K} \tag{3.6}$$

Where $1/SSE_k$ is the inverse reconstructive error on category channel $k$ from the set of categories $K$.

Exemplar models make classification decisions based on associations between category labels and regions of stimulus space. In contrast, DIVA posits that classification decisions are the result of deciding which category-specific description of a stimulus best matches the model's experience. Importantly, DIVA has had relative success fitting human behavior on a number of classification learning problems (Kurtz, 2007; Conaway & Kurtz, 2017), and has made novel, empirically supported predictions about stimulus generalization that contrast with predictions made by exemplar models (Conaway, 2016). This makes DIVA an ideal candidate model for the present work.

Like an autoencoder, DIVA's hidden layer represents stimuli as statistical relationships between their features. Using DIVA's hidden representational space as a predictor of human similarity ratings leads to interesting predictions about why categorical perception effects might emerge. If categories can be described by the statistical relationships between stimulus features, then stimuli from the same category (or region of space) should be initially compressed perpendicular to the most prominent component that describes the data (3.2 shows how this behavior of the model can lead to an increase in within-category similarity during learning)[4]. This is in contrast with the typical intuition that categorical percep-

[4]This mechanism might also explain why stimuli from different categories are rated as *less* similar through learning (though this is likely sensitive to the particular category structure being learned).

tion reflects a functional mechanism that makes categories easier to distinguish (Goldstone, 1994b; Pothos & Reppa, 2014). Rather, categorical perception occurs in DIVA as a result of forming compressed representations of stimuli (not necessarily along category boundaries). In summary, DIVA provides a fundamentally distinct explanation of categorical perception effects, making it an ideal predictor of similarity ratings in the context of the present study.

### 3.1.3 Theory Of Task

Categorical perception effects emerge in ALCOVE through attentional allocation towards diagnostic features (figure 3.1); this occurs as a result of optimizing ALCOVE to correctly classify presented stimuli (i.e., a discriminative learning objective). Categorical perception effects emerge in DIVA through compression along statistical components (figure 3.2); this occurs as a result of optimizing DIVA's weights to correctly infer the features of presented stimuli (i.e., generative learning objective). Given that categorical perception emerges in both models with entirely different learning objectives, one might expect that the manifestation of category perception in human learners is also mediated by the task (or, mode) in which categories are learned. The present work proposes a novel model of categorization, the *Theory-of-Task* model (ToT), that aims to provide a multifaceted account of categorical perception effects in human learners.

A multilayer neural network classifier, or *multilayer-perceptron* (MLP), consists of a number of layers that map stimuli to category labels; each layer is a set of weighted connections that transforms incoming data into some new representational space. This is structurally similar to the autoencoder architecture described earlier, but with a discriminative learning objective. During learning,

the hidden representational space projects stimuli into a new space where they are easier to discriminate. In the extreme case, this can result in maximal expansion between categories and maximal compression within categories[5] (figure 3.3). While this is particularly adaptive for classification, it might also reduce the models sensitivity to important, structural information about the category structure (for example, within-category correlations). Evidence of learners' sensitivity to generative properties of categories (Levering & Kurtz, 2015; Chin-Parker & Ross, 2004) suggests that a complete account of human category representations should retain generative properties of categories to some degree. This serves as the motivating principle behind the ToT architecture.

In the ToT architecture, stimuli are represented as vectors of feature values (similar to ALCOVE & DIVA). This information is then fed through weighted connections to the model's hidden layer. This step is identical to standard MLPs and Autoencoders. Critically, the activation of the hidden layer is propagated to two channels simultaneously, a classification channel and inference channel (which are identical to the output layers of an MLP and DIVA, respectively). The weights of the model are optimized to jointly predict both the features and the category label of a presented stimulus, given the following error function:

$$Error = (1-d)\sum(a_c - t_c)^2 + d\sum(a_i - t_i)^2 \tag{3.7}$$

where $a_c$ and $a_i$ are the output activations of the classification and inference channels, and $t_c$ and $t_i$ are vectors representing the category label and feature

[5]qualitatively similar to the categorical perception effects typically reported in the literature (Tijsseling & Harnad, 1997)

values of the *target* stimulus (respectively). The $d$ parameter controls the discriminative bias of the model; that is, a higher $d$ value will more heavily weigh the discriminative learning objective, while a lower value will more heavily weigh the models inference learning objective.

The ToT architecture makes a number of critical predictions about human category learning. Most notably, ToT predicts that human category representations are shaped by a combined learning objective of category diagnosis and category understanding (e.g., classification & feature inference, respectively); this results in a hidden representation that expands the distance between categories and compresses the distance within categories without completely masking structural properties of the categories themselves (see figure 3.3). Another prediction of the ToT model is that the category label of a stimulus is treated as another feature to be predicted (similar to the RATIONAL model of categorization; Anderson, 1991)[6]. ToT predicts that any difference in learning between category labels and category features results only from the inherent statistical properties of the category labels themselves[7] (in the context of the present experiment, category labels are features that are perfectly predictive and mutually exclusive).

## 3.2 Simulation Methods

DIVA, ALCOVE, & ToT have a number of hyperparameters that greatly impact their behavior. The hyperparameters for each model were selected using a random search of over 10,000 possible sets of hyperparameter values. For each

[6]though see Markman and Ross (2003) for a critique of this design choice in cognitive modeling

[7]in addition to the discriminative bias parameter built into the model

candidate hyperparameter set, models were trained for 7 blocks of randomized stimulus presentations[8], matching the learning experience of participants. Stimuli were coded as a vector of feature values. Models were then queried according to how well they fit the learning acquisition data as well as the similarity ratings provided by human participants.

### 3.2.1 Assessing Psychological Plausibility

Prior research has suggested that the representational space of connectionist models might provide promising avenue for explaining categorical perception effects in humans (Peterson et al., 2016; Tijsseling & Harnad, 1997; Khaligh-Razavi & Kriegeskorte, 2014). Following this line of research, the present work investigates whether the representational space of existing models of categorization provide adequate predictions about categorical perception effects in the context of a TACL experiment. Typically, leading models of categorization have been used to explain 2 key aspects of human behavior: (1) learning acquisition rates and (2) stimulus generalization. Following this tradition, models were first evaluated according to how well they fit the learning acquisition rates of the 3 categories averaged across all participants in each condition (using sum squared error as a fit metric; figure 3.4).

Once models were queried for their best fit to the learning data, they were then additionally assessed according to how well their representational space predicted human pair-wise similarity judgments averaged across all participants in each condition. Querying the representational space of connectionist models is relatively

[8]in each block, all stimuli in the training set are presented once

straight forward. Each stimulus in the present experiment can be described by a set of features with continuous values. In standard connectionist models, these feature values are represented in a new geometric space determined by the hidden nodes of the network. The distance between two stimuli in this new geometric space provides a metric of *similarity*. This distance has been used to successfully predict behavioral and physiological responses to categorical stimuli (Peterson et al., 2016; Lake et al., 2015; Khaligh-Razavi & Kriegeskorte, 2014), making it an ideal predictor of similarity judgments in the context of this study.

By tethering the models by their fits to learning data, the present work provides a critical challenge to existing theories on human categorization; this helps overcome the existing difficulty in falsify existing models fit to learning data alone. In addition, this particular analysis will shed light on whether or not their is a relationship between learning acquisition data and the representational space utilized by models fit to learning acquisition data. Should there be a close correspondence between fits to learning acquisition data and similarity judgments, the present work might provide a compelling benchmark challenge for new and existing theories. Alternatively, a lack of correspondence between these two metrics might suggest a severe limitation in existing models of category learning regarding their ability to provide a wholistic account of the many empirical phenomena that manifest themselves in TACL experiments.

### Training Models in Different Learning Modes

To the author's knowledge, previous studies have not looked into whether ALCOVE and DIVA's hidden representational space provide adequate predictions

of human similarity judgments. In addition, neither model has been extended to learning modes in categorization experiments other than classification. Another goal of the present work is to determine whether existing models can be extended to the endorsement learning mode in the present study.

The endorsement learning mode differs from classification in a critical way. Recall that in classification learning, subjects are asked to correctly identify which of 3 provided category labels matches the provided stimulus; subjects are then told the correct category that the stimulus belongs to. Consequently, each trial provides subjects with complete information about how the stimulus relates to each of the 3 provided categories. Computational modeling typically represents this feedback in the form of a one-hot coded vector.

In endorsement learning, subjects are asked to correctly identify whether a provided stimulus matches a provided category label. They are then given feedback on whether or not their endorsement was correct. This results in 2 qualitatively different forms of feedback (figure 2.3). When the provided label matches the stimulus, subjects are provided complete feedback about category membership. In contrast, when the provided category label does not match the stimulus, subjects are only provided feedback regarding the provided category label (feedback regarding the remaining 2 category labels is ambiguous). To simulate this effect in computational models, *model weights were only updated when a randomly provided category label matched the true category label of the provided stimulus.*

### 3.2.2 Querying Representational Space

ALCOVE implicitly predicts that categorical perception effects arise from changes in attentional allocation towards task-relevant features of stimuli. When

assessing the representational similarity between two stimuli, this equates to calculating the Minkowski (i.e., spatial) distance between the features of two stimuli and then weighting those distances according to ALCOVE's learned attentional weights; this is given by the equation:

$$s_{ij} = e^{-c((|i-j| \cdot W_\alpha)^r)^{1/r}} \tag{3.8}$$

where $i$ and $j$ are the feature vectors of the two stimuli being compared.

The representational spaces of DIVA and ToT can be computed in a similar way; however, instead of using the original features of stimuli, the representational space is defined by the activity of all of the nodes in the hidden layer. The similarity of two items $i$ and $j$ will be defined as the euclidean distance in DIVA's representational space, given by the equation:

$$s_{ij} = (\sum (h_i - h_j)^2)^{1/2} \tag{3.9}$$

where $h$ is a vector of the activations of all nodes in the model's hidden layer. Once calculated, the pairwise distances for ALCOVE, DIVA, and ToT were scaled from 0 - 1 (to match the range used by human subjects). Following the procedure of prior work assessing the psychological plausibility of connectionist representational space (Long et al., 2018), the fit to human similarity ratings is determined by the Kendall rank-order correlation between the pairwise distance in a model's representational space and the pairwise similarity ratings of human learners.

## 3.3 Results & Discussion

To assess psychological plausibility, models were first fit to the behavioral learning data of the training phase of the experiment. Then, models were assessed according to how well their hidden layer representations predicted the pairwise similarity ratings of classification and endorsement learners. For the classification condition, DIVA provided the best relative fit to the learning data of the 3 category structures ($MSE$ = 0.0024), followed closely by ALCOVE ($MSE$ = 0.003). For the endorsement condition, ALCOVE provided the best relative fit to the learning data ($MSE$ = 0.0068), followed closely by DIVA ($MSE$ = 0.0079). The ToT model provided relatively poor fits to the learning data in both the classification ($MSE$ = 1.445) and endorsement ($MSE$ = 0.832) conditions. For the similarity predictions of the models fit to the learning data, ALCOVE produced the best fit to the similarity ratings in the classification condition ($\tau$=.82), followed by DIVA ($\tau$=.72) and ToT ($\tau$=.72). For the endorsement condition, ALCOVA again produced the best fit ($\tau$=.76), followed closely by DIVA ($\tau$=.74), and then ToT ($\tau$=.70)[9].

Both DIVA and ALCOVE provided comparably good fits to the learning data for the classification and endorsement learning modes, while the ToT model provided relatively poor fits to the learning data. When models were fit to the learning data, ALCOVE provided the best fit to the similarity ratings in the classification condition, while ALCOVE and DIVA provided comparable fits to the endorsement condition. ToT provided the relatively least plausible account to the similarity ratings. Collectively, the variance of similarity ratings accounted for by

[9]Note: the same qualitative pattern of results emerges when querying models for their fits to similarity ratings in isolation (as opposed to tethering the models learning data first)

the models ranged from 48%-67%. Interestingly, the standard exponentiated featural distance of the stimulus pairs explained 67%-69% of the variance in the data. Because ALCOVE's representational space is computed by taking the attention-modified exponentiated featural distance between stimuli, it is expected that its ability to account for the data would highly correspond to the actual distances between stimuli. However, DIVA's success at predicting similarity ratings comes as a result of trying to capture statistical components of the data, with no particular assumptions about the representational space itself.

| **Classification Learning** | | | |
|---|---|---|---|
| **Model** | **Sample Size** | **SSE (Learning Acquisition Rate)** | **Tau (Similarity Ratings)** |
| **ALCOVE** | 10000 | 0.003 | .82 |
| **DIVA** | 10000 | 0.0024 | .72 |
| **ToT** | 10000 | 1.445 | .72 |

| **Endorsement Learning** | | | |
|---|---|---|---|
| **ALCOVE** | 10000 | 0.0068 | .76 |
| **DIVA** | 10000 | 0.0079 | .74 |
| **ToT** | 10000 | 0.832 | .70 |

Table 3.1: Psychological plausibility metrics for each model after being fit to the learning data of the experiment.

| **Classification** | | | | | |
|---|---|---|---|---|---|
| **ALCOVE** | | **DIVA** | | **Theory of Task** | |
| **Specificity** | 2.9 | **Learning Rate** | .24 | **Learning Rate** | .05 |
| **Attention Learning Rate** | .11 | **Hidden Nodes** | 14 | **Hidden Nodes** | 6 |
| **Association Learning Rate** | .17 | **Beta Value** | 1.6 | **Discriminative Bias** | .95 |
| **Response Mapping** | 2.4 | | | | |

Table 3.2: Hyperparameters for each model that produced the best psychological fits to the classification condition.

| Endorsement | | | | | |
|---|---|---|---|---|---|
| **ALCOVE** | | **DIVA** | | **Theory of Task** | |
| **Specificity** | 13.7 | **Learning Rate** | .24 | **Learning Rate** | .06 |
| **Attention Learning Rate** | 1.9 | **Hidden Nodes** | 8 | **Hidden Nodes** | 3 |
| **Association Learning Rate** | .81 | **Beta Value** | 1.3 | **Discriminative Bias** | .98 |
| **Response Mapping** | 4.7 | | | | |

Table 3.3: Hyperparameters for each model that produced the best psychological fits to the endorsement condition.

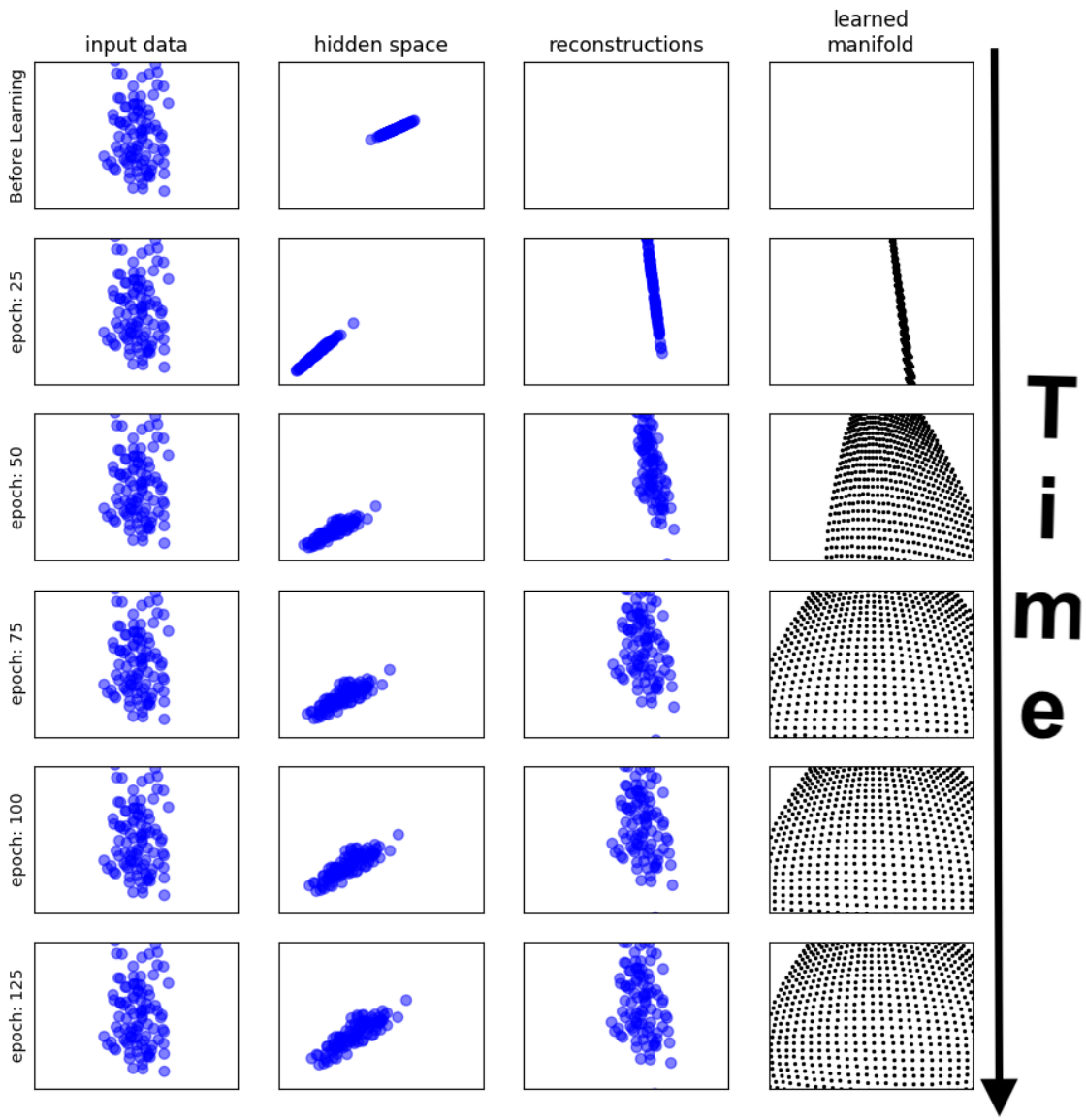


Figure 3.2: Visualization of the hidden and reconstructive layer of an autoencoder with 2 hidden nodes. In this simulation, data were first reconstructed along the strongest statistical component; this provides one route through which within-category compression effects could occur.

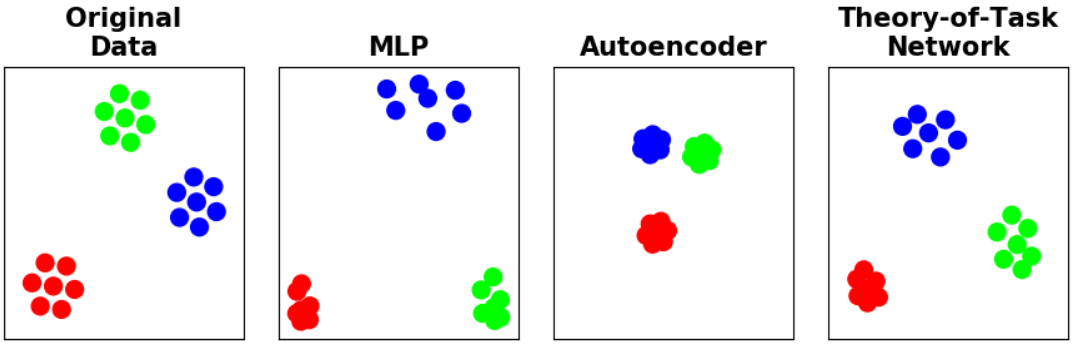


Figure 3.3: Visualization of the hidden representational space of an MLP, and autoencoder, and the ToT model. Unlike an autoencoder, the hidden space of an MLP compresses and expands the category structure in a way that highly distorts the original structure of the data. As predicted, the ToT model compresses and expands the category structure while still retaining the core structure of the original data.

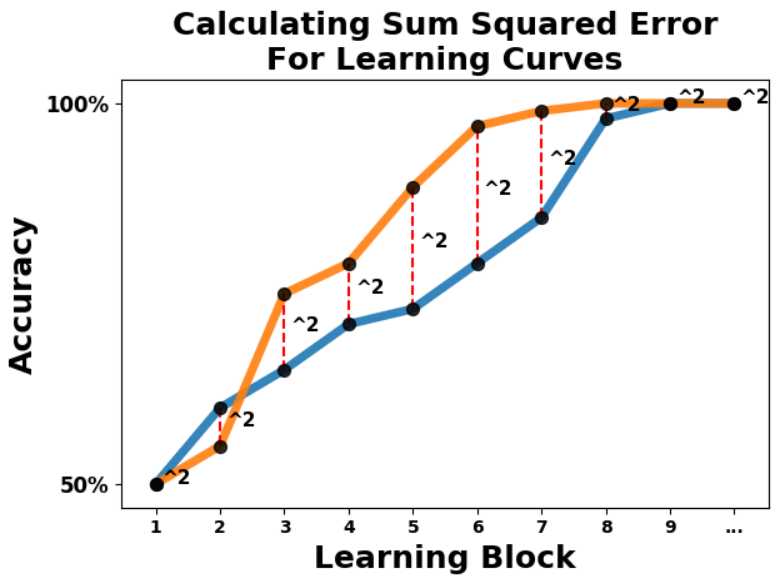


Figure 3.4: Fits to learning data were defined by the sum squared error between human accuracy and model accuracy at each training block.

# 4 General Discussion

The present experiment had 2 primary objectives: (1) demonstrate the impact of learning mode on categorical perception effects[1], and (2) explore whether existing computational models could provide psychologically plausible explanations of categorical perception effects in human learners. To this end, subjects learned category structures consisting of 3 categories of 2-dimensional stimuli, and were then asked to provide similarity ratings between pairs of stimuli. Categorical perception effects were operationalized as differences in similarity ratings between subjects who engaged in a category learning task (endorsement or classification) and subjects that did not. The psychological plausibility of models was determined by how well the models could predict the structural patterns of similarity ratings after being fit to the learning acquisition rate of subjects.

The behavioral results provide a partial replication of the categorical perception effects typically observed in the literature. Namely, both endorsement learning and classification learning resulted in less between-category similarity ratings than controls. This is contrary to the original hypothesis that endorsement learning would lead to less discriminative category representations (from which we wouldn't expect categorical perception effects to occur). Importantly, there was no evidence that within-category similarity ratings differed as a function of learn-

[1]particularly the phenomena of between-category expansion & within-category compression

ing. This might reflect a lack of robustness of the within-category compression effect. Another seemingly likely explanation is that within-category compression effects were masked by a ceiling effect of within-category stimulus pairs being very similar to each other. This same effect may have also masked any differences in the learning acquisition rate between category structures[2]. The category structures in this experiment were learned relatively quickly compared to benchmark structures (Nosofsky, Gluck, et al., 1994; Kruschke, 1993); this makes sense given that none of the structures were overlapping and within-category exemplars were clustered very close together. Prior work suggests that categorical perception effects are less likely to occur when the category learning task is relatively easy (Pothos & Reppa, 2014). One remedy to the ceiling effects observed with these category structures is to expand the distance of within-category clusters. Preliminary pilot data suggests that this decreases the acquisition rate of the category structures, and may reveal meaningful differences between the 3 category structures utilized in this experiment.

### Categorical Perception in Endorsement Learning

The finding that endorsement learning produced between-category expansion effects was contrary to the original hypothesis that endorsement learning would result in acquisition of less discriminative category representations in human learners. One explanation is that endorsement learning biases subjects towards learning the same discriminative properties of categories as classification learning. Al-

[2]Analysis of learning curves may have also been limited by the metric used (first accurate block scores). Future work could test learning acquisition data using growth curve analysis using logistic models to represent learning over time (McArdle & Nesselroade, 2003)

though the specific task of subjects in endorsement learning is different from classification learning (yes/no decisions as opposed to category label selection), the two learning modes share critical similarities. For example, in both endorsement and classification learning, category membership is exclusive (a stimulus can only be a member of 1 category). Like classification, learning what *doesn't* describe a category can sometimes be just as valuable as learning what *does* describe a category[3]. In may be that the only critical difference between between endorsement learning and classification learning is *the informational content of the feedback provided during each trial of training.* The finding that endorsement learning takes longer to master can be taken as evidence supporting this conclusion. Future work might benefit from utilizing more well established learning modes that are known to alleviate diagnostic bias (such as inference or observational learning; Levering & Kurtz, 2015; Chin-Parker & Ross, 2004).

If endorsement learning utilizes the same underlying cognitive processes as classification learning, it would be expected that the aggregate generalization profiles of learners during the unsupervised test phase would be similar in the endorsement and classification conditions. For the 0-Share and 1-Share conditions, this appears to be the case. However, for the 2-Share condition, the aggregate generalization profiles of learners differ in a way that *suggests* limited attentional bias for endorsement learners (unlike classification learners, some of whom appear to be using multi-faceted attentional solutions to solve the problem). Interestingly, this may suggest that endorsement learning encourages discriminative strategies to a

[3]The use of 3 categories was intended to reduce the likelihood that subjects would focus purely diagnostic features of category membership, but it does not completely eliminate the possibility that subjects utilize partially predictive, discriminative knowledge to solve the task.

lesser degree than classification learning. One particular limitation of using an endorsement task is that it does not provide a complete generalization profile for each subject, making this particular question difficult to answer. This limitation could be remedied by using a classification task for generalization after using an endorsement task for training.

### The Adaptive Role of Representational Shifts

One key finding of the present work is that differences in between-category similarity ratings across the learning and control conditions was mediated by the featural distance of between-category stimulus pairs. That is, there were stronger between-category expansion effects when between-category stimulus pairs were very similar (i.e., more difficult to discriminate). This suggests that categorical perception effects play an adaptive role during human category learning. It also has a major implication regarding whether or not categorical perception effects reflect a true change in the perception of stimuli (or instead just a strategic response bias Goldstone et al., 2001). It is possible that subjects explicitly utilize categorical information when making similarity judgments (i.e., stimuli are rated as less similar *because they belong to different categories*, not because of any perceived change in their appearance). However, it is unclear how the strategic bias account would predict the impact of featural distance on between-category expansion effects, or how the use of featural distance to modulate similarity ratings would serve any strategic goal of the learner. While it is difficult to completely discount the strategic bias account of categorical perception effects, the correlation between featural distance and between-category expansion suggests a functional,

implicit role of the categorical perception phenomena.

### Psychological Plausibility of Existing Computational Models

Existing computational models of categorization were assessed according to their ability to predict the pairwise similarity ratings of human subjects after being fit to the learning acquisition rate of the 3 category structures. DIVA and ALCOVE provided comparable fits to the learning acquisition rates, while ALCOVE slightly outperformed DIVA in predicting similarity ratings. Interestingly, inverse-exponential distance between stimuli in feature space (Shepard, 1987) was the best predictor of similarity ratings, despite having fewer parameters than existing computational models[4]. The ToT model in its present form provided the worst relative fit to the behavioral data.

ALCOVE's predictive success suggests that subjects are utilizing an exemplar-learning strategy when solving the 3 category structures utilized in the present experiment. However, ALCOVE utilizes a number of empirically-inspired design choices that aren't necessarily exclusive to an exemplar theory of category learning (Murphy, 2004). For example, ALCOVE utilizes an attention learning process that biases task-relevant features. There are a number of empirical phenomena that suggest humans can utilize an attention learning mechanism during category learning (Kruschke, 2011). For example, when categories can be diagnosed by a linearly-separable boundary in feature space, subjects learn faster when that boundary is defined by a single dimension (as opposed to multiple features; Kruschke, 1993). Additionally, the classic work by Shepard et al. (1961) suggests

[4]The inverse-exponential distance function has only one parameter: specificity.

that the rate of category acquisition decreases as the number of relevant features increases. DIVA and ToT in their current form do not utilize any sort of explicit attention learning mechanism; the implementation of a task-relevant attentional mechanisms might improve DIVA and ToT's ability to predict the similarity ratings in the present experiment[5]. Another empirically-driven design choice in ALCOVE is the use of an inverse-exponential similarity space. This is inspired by Shepard's (1987) discovery that human stimulus generalization can be described by an inverse-exponential function. Importantly, inverse-exponential distance was the best predictor of human similarity judgments in the present experiment. Like attention, DIVA and ToT might benefit from utilizing an inverse-exponential function in the manifold that describes their representational space[6].

The goal of the present work was to demonstrate the impact of learning mode on categorical perception effects, and to see whether connectionist models predict the particular pattern of results. Connectionist models were specifically chosen because of their propensity to demonstrate categorical perception effects during learning (Tijsseling & Harnad, 1997). One meaningful limitation of the present work is that all models were only fit to similarity ratings of the category learning conditions *after training*. While the prediction of similarity ratings post-learning is a meaningful challenge for existing models, it doesn't directly provide evidence that models manifest categorical perception effects in a way that's similar to hu-

[5]Selective attention processes fit naturally within the framework of connectionist models, and have recently seen a wave of success in the machine learning literature (particularly in the domain of language; Firat, Cho, & Bengio, 2016; Bahdanau, Chorowski, Serdyuk, Brakel, & Bengio, 2016).

[6]Interestingly, Gaussian Mixture Models of human categorization take advantage of inverse-exponential distance without sacrificing any particular tendency to learn generative properties of data

man learners. While this could be remedied by having subjects engage in a similarity rating task *before* and *after* learning, prior work suggests that pre-exposure to stimuli influences later learning (Folstein et al., 2010). However, this confounding design choice might be necessary to directly investigate whether computational models can predict categorical perception effects in human learners.

### Different Routes to Categorical Perception Effects

Like DIVA & ToT, ALCOVE predictions of structural patterns of similarity ratings were based on the activations of its hidden layer. ALCOVE's hidden layer is an inverse-exponential distance function that biases task-relevant features (i.e., attention); this has the effect of stretching and compressing the featural space in a way that increases the distance between categories along diagnostic boundaries. This is one way in which the between-category expansion effect typically reported in the literature might occur. Interestingly, this also suggests that within-category compression effects *should only occur when within-category clusters are spaced along nondiagnostic dimensions* that get compressed during ALCOVE's attention learning process. While the category structures utilized in this experiment don't lend themselves directly to this question, future work could directly test ALCOVE predictions using category structures where within-category clusters vary along diagnostic or nondiagnostic dimensions exclusively (in addition to category structures where there is no attentional solution; figure 4.1).

DIVA provides a very different explanation of why categorical perception effects might emerge during classification learning. DIVA's classification decisions are determined by comparing which category-specific statistical model best de-

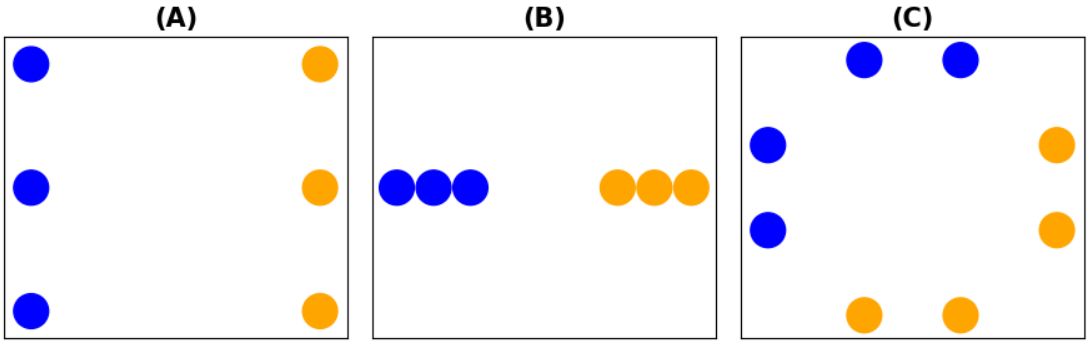


Figure 4.1: 3 possible category structures for exploring categorical perception in ALCOVE. In structure (A), ALCOVE's learned attentional weights should compress categories along the nondiagnostic dimension. In structure (B), stimuli within a category do not vary along nondiagnostic dimensions (i.e., categorical perception effects should not occur). In structure (C), there is no attentional solution to solve the category structure (from: Kruschke, 1993)

scribes an experienced stimulus. The hidden representational space of DIVA is driven to learn generative (rather than discriminative) properties of the categories (Kurtz, 2007). In doing so, stimuli are initially compressed orthogonal to pronounced statistical components in feature space[7]. As a result, DIVA predicts that categorical perception effects should occur primarily when the statistical components of stimuli interact with the geometry of the category structure being learned (figure 4.2). DIVA also makes specific predictions about the time course of categorical perception effects, which might require querying subjects similarity ratings at multiple stages throughout training.

### Conclusions

The present work demonstrates that categorical perception effects occur in both classification learning and endorsement learning. It also demonstrates a re-

[7]While preliminary simulations suggest this effect occurs when stimuli are described by continuous features, it is unclear whether this property extends to stimuli described by binary or discrete features.

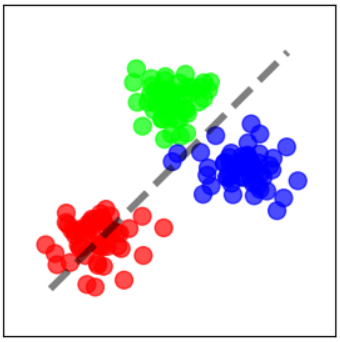

Figure 4.2: DIVA makes specific predictions about the way stimuli are compressed and expanded in stimulus space. During learning, stimuli are compressed orthogonal to the statistical component that best captures variability in the data (represented here as the black dashed line); as a result, categorical perception effects are sensitive to the way in which the categories are arranged.

lationship between pairwise stimulus distance and the magnitude in which categorical perception effects occur. While none of the existing models were as predictive of the structural pattern of similarity ratings as an inverse-exponential distance function, ALCOVE provided the best fit relative to the 3 candidate models that were tested. The overall findings from the present work demonstrate that categorical perception is a fairly robust phenomenon, and highlight what might be a bidirectional relationship between the perception and categorization mechanisms that serve human cognition.

# References


Murphy, G. (2004). The big book of concepts. MIT press.

Smith, E. E., & Medin, D. L. (1981). *Categories and concepts*. Harvard University Press Cambridge, MA.

Wittgenstein, L. (1953). Philosophical investigations.(gem anscombe, trans.) oxford. *UK: Blackwell*.

Rosch, E. (1975). Cognitive representations of semantic categories. *Journal of experimental psychology: General*, *104*(3), 192.

Rosch, E., & Mervis, C. B. (1975). Family resemblances: Studies in the internal structure of categories. *Cognitive psychology*, *7*(4), 573–605.

Nosofsky, R. M. (1986). Attention, similarity, and the identification–categorization relationship. *Journal of experimental psychology: General*, *115*(1), 39.

Ashby, F. G., & Lee, W. W. (1991). Predicting similarity and categorization from identification. *Journal of Experimental Psychology: General*, *120*(2), 150.

Kruschke, J. K. (1992). Alcove: An exemplar-based connectionist model of category learning. *Psychological review*, *99*(1), 22.

Kurtz, K. J. (2007). The divergent autoencoder (diva) model of category learning. *Psychonomic Bulletin & Review*, *14*(4), 560–576.

Love, B. C., Medin, D. L., & Gureckis, T. M. (2004). Sustain: A network model of category learning. *Psychological review*, *111*(2), 309.

Anderson, J. R. (1991). The adaptive nature of human categorization. *Psychological review*, *98*(3), 409.

Markman, A. B., & Ross, B. H. (2003). Category use and category learning. *Psychological bulletin*, *129*(4), 592.

Levering, K. R., & Kurtz, K. J. (2015). Observation versus classification in supervised category learning. *Memory & cognition*, *43*(2), 266–282.

Shepard, R. N., Hovland, C. I., & Jenkins, H. M. (1961). Learning and memorization of classifications. *Psychological monographs: General and applied*, *75*(13), 1.

Bruner, J., Goodnow, J., & Austin, G. (1956). A study of thinking john wiley & sons. *Ine, New York*.

Wills, A. J., & Pothos, E. M. (2012). On the adequacy of current empirical evaluations of formal models of categorization. *Psychological bulletin*, *138*(1), 102.

Kurtz, K. J. (2015). Human category learning: Toward a broader explanatory account. In *Psychology of learning and motivation* (Vol. 63, pp. 77–114). Elsevier.

Nosofsky, R. M., Gluck, M. A., Palmeri, T. J., McKinley, S. C., & Glauthier, P. (1994). Comparing modes of rule-based classification learning: A replication and extension of shepard, hovland, and jenkins (1961). *Memory & cognition*, *22*(3), 352–369.

Kruschke, J. K. (1993). Human category learning: Implications for backpropagation models. *Connection Science*, *5*(1), 3–36.

Erickson, M. A., & Kruschke, J. K. (1998). Rules and exemplars in category learning. *Journal of Experimental Psychology: General*, *127*(2), 107.

Witnauer, J. E., Urcelay, G. P., & Miller, R. R. (2014). The error in total error reduction. *Neurobiology of learning and memory*, *108*, 119–135.

Shepard, R. N. (1987). Toward a universal law of generalization for psychological science. *Science*, *237*(4820), 1317–1323.

Kang, K., Shapley, R. M., & Sompolinsky, H. (2004). Information tuning of populations of neurons in primary visual cortex. *Journal of neuroscience*, *24*(15), 3726–3735.

Butts, D. A., & Goldman, M. S. (2006). Tuning curves, neuronal variability, and sensory coding. *PLoS biology*, *4*(4), e92.

Guttman, N., & Kalish, H. I. (1956). Discriminability and stimulus generalization. *Journal of experimental psychology*, *51*(1), 79.

Minda, J. P., & Smith, J. D. (2001). Prototypes in category learning: The effects of category size, category structure, and stimulus complexity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *27*(3), 775.

Palmeri, T. J. (1999). Learning categories at different hierarchical levels: A comparison of category learning models. *Psychonomic Bulletin & Review*, *6*(3), 495–503.

Nosofsky, R. M., & Johansen, M. K. (2000). Exemplar-based accounts of" multiple-system" phenomena in perceptual categorization. *Psychonomic Bulletin and Review*, *7*(3), 375–402.

Nosofsky, R. M., & Palmeri, T. J. (1998). A rule-plus-exception model for classifying objects in continuous-dimension spaces. *Psychonomic Bulletin & Review*, *5*(3), 345–369.

Ashby, F. G., & Gott, R. E. (1988). Decision rules in the perception and categorization of multidimensional stimuli. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *14*(1), 33.

Ashby, F. G., Alfonso-Reese, L. A., Waldron, E. M., et al. (1998). A neuropsychological theory of multiple systems in category learning. *Psychological review*, *105*(3), 442.

Livingston, K. R., Andrews, J. K., & Harnad, S. (1998). Categorical perception effects induced by category learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *24*(3), 732.

Goldstone, R. L. (1994a). The role of similarity in categorization: Providing a groundwork. *Cognition*, *52*(2), 125–157.

Liberman, A. M., Harris, K. S., Hoffman, H. S., & Griffith, B. C. (1957). The discrimination of speech sounds within and across phoneme boundaries. *Journal of experimental psychology*, *54*(5), 358.

Harnad, S. (1987). Psychophysical and cognitive aspects of categorical perception: A critical overview. In *Categorical perception: The groundwork of cognition* (pp. 1–52). Cambridge University Press.

Goldstone, R. L. (1994b). Influences of categorization on perceptual discrimination. *Journal of Experimental Psychology: General*, *123*(2), 178.

Goldstone, R. L., Lippa, Y., & Shiffrin, R. M. (2001). Altering object representations through category learning. *Cognition*, *78*(1), 27–43.

Gureckis, T. M., & Goldstone, R. L. (2008). The effect of the internal structure of categories on perception. In *Proceedings of the 30th annual conference of the cognitive science society* (pp. 1876–1881). Austin, TX.

Folstein, J. R., Gauthier, I., & Palmeri, T. J. (2010). Mere exposure alters category learning of novel objects. *Frontiers in psychology*, *1*, 40.

Pothos, E. M., & Reppa, I. (2014). The fickle nature of similarity change as a result of categorization. *Quarterly Journal of Experimental Psychology*, *67*(12), 2425–2438.

Jäkel, F., Schölkopf, B., & Wichmann, F. A. (2008). Similarity, kernels, and the triangle inequality. *Journal of Mathematical Psychology*, *52*(5), 297–303.

Tversky, A., & Gati, I. (1982). Similarity, separability, and the triangle inequality. *Psychological review*, *89*(2), 123.

Goldstone, R. L., & Son, J. Y. (2005). Similarity. In *The oxford handbook of thinking and reasoning*.

Podgorny, P., & Garner, W. (1979). Reaction time as a measure of inter-and intraobject visual similarity: Letters of the alphabet. *Perception & Psychophysics*, *26*(1), 37–52.

Holyoak, K. J., & Gordon, P. C. (1983). Social reference points. *Journal of Personality and Social Psychology*, *44*(5), 881.

Nosofsky, R. M., Palmeri, T. J., & McKinley, S. C. (1994). Rule-plus-exception model of classification learning. *Psychological review*, *101*(1), 53.

McClelland, J. L., Rumelhart, D. E., Group, P. R., et al. (1987). *Parallel distributed processing*. MIT press Cambridge, MA:

Peterson, J. C., Abbott, J. T., & Griffiths, T. L. (2016). Adapting deep network features to capture psychological representations. *arXiv preprint arXiv:1608.02164*.

He, K., Zhang, X., Ren, S., & Sun, J. (2015). Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *Proceedings of the ieee international conference on computer vision* (pp. 1026–1034).

Tijsseling, A., & Harnad, S. (1997). Warping similarity space in category learning by backprop nets. In *Proceedings of simcat 1997: Interdisciplinary workshop on similarity and categorization* (pp. 263–269). Department of Artificial Intelligence, Edinburgh University.

Long, B., Yu, C.-P., & Konkle, T. (2018). Mid-level visual features underlie the high-level categorical organization of the ventral stream. *Proceedings of the National Academy of Sciences*, *115*(38), E9015–E9024.

Lake, B. M., Zaremba, W., Fergus, R., & Gureckis, T. M. (2015). Deep neural networks predict category typicality ratings for images. In *Cogsci*.

Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis-connecting the branches of systems neuroscience. *Frontiers in systems neuroscience*, *2*, 4.

Laakso, A., & Cottrell, G. (2000). Content and cluster analysis: Assessing representational similarity in neural systems. *Philosophical psychology*, *13*(1), 47–76.

Khaligh-Razavi, S.-M., & Kriegeskorte, N. (2014). Deep supervised, but not unsupervised, models may explain it cortical representation. *PLoS computational biology*, *10*(11), e1003915.

Lee, H., Grosse, R., Ranganath, R., & Ng, A. Y. (2009). Convolutional deep belief networks for scalable unsupervised learning of hierarchical representations. In *Proceedings of the 26th annual international conference on machine learning* (pp. 609–616). ACM.

Baker, N., Lu, H., Erlikhman, G., & Kellman, P. J. (2018). Deep convolutional networks do not classify based on global object shape. *PLoS computational biology*, *14*(12), e1006613.

Chin-Parker, S., & Ross, B. H. (2004). Diagnosticity and prototypicality in category learning: A comparison of inference learning and classification learn-

ing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *30*(1), 216.

Kattner, F., Cox, C. R., & Green, C. S. (2016). Transfer in rule-based category learning depends on the training task. *PloS one*, *11*(10), e0165260.

Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., ... Lindeløv, J. K. (2019). Psychopy2: Experiments in behavior made easy. *Behavior research methods*, *51*(1), 195–203.

Medin, D. L., & Schaffer, M. M. (1978). Context theory of classification learning. *Psychological review*, *85*(3), 207.

Rosseel, Y. (2002). Mixture models of categorization. *Journal of Mathematical Psychology*, *46*(2), 178–210.

Kramer, M. A. (1991). Nonlinear principal component analysis using autoassociative neural networks. *AIChE journal*, *37*(2), 233–243.

Conaway, N., & Kurtz, K. J. (2017). Similar to the category, but not the exemplars: A study of generalization. *Psychonomic bulletin & review*, *24*(4), 1312–1323.

Conaway, N. (2016). *Re-evaluating the reference point view of human classification learning* (Doctoral dissertation, State University of New York at Binghamton).

McArdle, J. J., & Nesselroade, J. R. (2003). Growth curve analysis in contemporary psychological research. *Handbook of psychology*, 447–480.

Kruschke, J. K. (2011). Models of attentional learning. *Formal approaches in categorization*, *120*.

Firat, O., Cho, K., & Bengio, Y. (2016). Multi-way, multilingual neural machine translation with a shared attention mechanism. *arXiv preprint arXiv:1601.01073*.

Bahdanau, D., Chorowski, J., Serdyuk, D., Brakel, P., & Bengio, Y. (2016). End-to-end attention-based large vocabulary speech recognition. In *2016 ieee international conference on acoustics, speech and signal processing (icassp)* (pp. 4945–4949). IEEE.